

Deliberation and Policy Outcomes: Evidence from the Textual Analysis of FOMC Transcripts *

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Abstract

Natural language processing is used to extract information from FOMC transcripts and construct quantitative text-based measures of voiced policy stance, emotions, and collaboration. These measures are inputs in an econometric model of deliberation where members interact with one another across rounds of a meeting and over time across meetings. Evidence shows that members learn from one another during within-meeting deliberation and exert influence across meetings. Although emotional tone has limited effects on policy stances and decisions, it has strong predictive power for dissent behavior.

JEL classification: D7, E5

Key Words: Collective decision-making, monetary committee, natural language processing, dissent, transparency.

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1 Introduction

Deliberation in collective decision-making bodies involves the thoughtful weighting of options by a group of individuals and the exchange of information and points of view among participants, possibly leading some of them to revise their initial stance. In the case of the Federal Open Market Committee (FOMC), monetary policy decisions are made by a group of 19 individuals—seven Governors of the Federal Reserve Board and 12 Reserve Bank Presidents, five of whom have voting rights—in regular meetings roughly six weeks apart. The relatively stable format of the meetings, with a specific proposal formulated by the chair and voted on at the end, and the availability of detailed transcripts make the FOMC an ideal body to study deliberation and its effect on policy outcomes.¹ Relying on recent advances in artificial intelligence (AI) and natural language processing (NLP), we extract information encoded in FOMC transcripts from March 1976 to December 2018 to construct quantitative text-based measures of policy stance, emotions, and collaboration that help us study how committee members interact with one another across rounds of a meeting and over time across meetings.

A number of theoretical papers study how committees aggregate information when members are allowed to deliberate, that is, to engage in pre-vote communication (e.g., Coughlan (2000), and Austen-Smith and Feddersen (2006)). This extension is especially relevant for the FOMC where members first share their private information by summarizing their assessment of current economic conditions in an “economic round,” and then express their views on the appropriate policy in a “policy round,” before the chair makes a proposal that is put to a vote. The empirical papers by Chappell et al. (2012) and Lopez-Moctezuma (2016) exploit the observation that policy recommendations are stated sequentially in the policy round and examine whether members learn from each other as they reveal their preferred interest rate. These authors identify the preferred rate through the manual reading of policy round transcripts but reach different conclusions as to whether a speaker influences subsequent ones.²

Our analysis of FOMC deliberation differs from the two above-mentioned papers in two respects.

¹Compared with legislatures and courts, the FOMC is also a simpler decision-making body in that the main decision concerns only one variable (say, the target for the federal funds rate), there are no side payments, and the identity of the agenda-setter (i.e., the chair) is stable and known.

²Chappell et al. (2012) focus on the chairmanship of Arthur Burns and find no evidence that members speaking later are influenced by those speaking earlier. Lopez-Moctezuma (2016) covers the longer period from 1970 to 2008 and finds that recommendations by early speakers predict the voiced policy preferences of subsequent speakers.

First, we focus on how deliberation evolves from the economic round to the policy round, rather than within the policy round itself. This is possible because text-analysis and transformer-based methods allow us to classify monetary policy stance already in the economic round, something that is not feasible with manual approaches. The advantage of studying across-round deliberation is that the economic and policy rounds map closely, respectively, to the pre-vote communication and voting stages in the theoretical literature on deliberation. Second, while Chappell et al. (2012) test for correlations in policy recommendations and Lopez-Moctezuma (2016) estimates a structural model of social learning, we estimate a reduced-form linear model of deliberation that, in addition to voiced policy stance, incorporates behavioral factors, such as the emotional tone of discussion (e.g., anger, happiness, fear) and the level of collaboration (measured by two personality traits, namely agreeableness and openness). The reduced form of our model is a vector autoregression (VAR) and standard procedures can be used to test whether the voiced policy stances by some members may be forecastable on the basis of the stances by other members or the tone of the discussion. While there is a growing literature examining the emotional tone of the FOMC chair in press conferences and congressional testimonies, and its effect on financial markets (e.g., see Gorodnichenko et al. (2023), Curti and Kazinnik (2023), and Alexopoulos et al. (2024)), to our knowledge, no paper has studied the mutual interaction between emotional tone, collaboration, and policy stance in FOMC deliberation and their joint effect on policy decisions and dissent behavior. In addition, this paper sheds light on an important question, namely whether 1) FOMC deliberation goes beyond self-contained presentations with members influencing each other within meetings, or 2) deliberation takes place over time, across meetings (see Schonhardt-Bailey and Bailey (2013) for a discussion).

Estimates of our model of deliberation indicate that voiced policy stances and emotional tone of board governors and bank presidents in the economic round are predictable on the basis of data from the Greenbooks. For instance, a rise in output growth increases the voiced stance for tighter policy, decreases the stance for looser policy, and reduces negative tone. In contrast, the chair remains circumspect in this round and his or her policy stance is not discernible at this stage. Estimates also show that policy stances by all members in the policy round are predictable on the basis of stances in the preceding economic round and the data, in a manner that suggests mutual influence across members. A key issue in theoretical models of deliberation is whether members are

willing to truthfully share their private information, especially when preferences are not aligned. Bayesian updating suggests that views contrary to a member’s expected bias are more credible and therefore more influential. Our results align with this prediction: we find that a stance for looser policy in the economic round by bank presidents—who are typically regarded as more hawkish than board governors—has a stronger influence on the policy stance of other members in the policy round and in the same direction. Likewise, a stance for tighter policy by governors has a greater effect in the following round and in the same direction. The general takeaway from these two results is that FOMC members learn from one another during within meeting deliberation and may be aware of each other’s biases, giving greater weight to opinions that are surprising relative to the source member’s usual stance.

Based on block-exogeneity tests, we find (perhaps, reassuringly) that emotional tone has limited or no effect on voiced policy stances and interest rate decisions. However, emotional tone has predictive power over dissent behavior with a more negative tone increasing the probability and rate of dissent and a more positive tone decreasing them. This result is important because previous research by Riboni and Ruge-Murcia (2014) shows that current dissents help forecasts future policy decisions by the FOMC. It is interesting to note that the increase in transparency in 1993—when the existence of transcripts to be made publicly available became known—is associated with a decrease in negative tone and increase in positive tone in both rounds, and a decrease in the probability and rate of dissent by board governors.

Estimates of the version of the model of long-term deliberation show that some of the effects above carry over across meetings. Although voiced preferences are persistent across meetings with autoregressive coefficients ranging from 0.13 to 0.32, other variables are also significant. Thus, we conclude that current policy stances are not mechanically determined by individual past stances but evolve as a result of deliberation. In particular, we find that bank presidents tend to be relatively more influential on the policy stances of members across meetings.

Our paper complements previous research that applies textual analysis to FOMC transcripts and other documents. Hansen et al. (2018) study how increased transparency after 1993 affected FOMC deliberations, making interventions more scripted and quantitative, and leading to greater conformity among less experienced members. The focus of this paper is different, as we examine how preferences are aggregated and how this process leads to changes in voiced policy stances. Using a

pre-defined dictionary, Shapiro and Wilson (2021) examine negative wording in FOMC transcripts to estimate the implicit inflation target, while Bordo et al. (2024) investigate the presence of wording related to commitment or discretion in FOMC deliberations. Laarits et al. (2025) use large language models to extract arguments from FOMC deliberations and find that members employ distinct frameworks to interpret common public information. A few papers have incorporated text-based measures into Taylor rules, demonstrating that they convey additional information and improve the forecast of policy decisions. Cieslak et al. (2023) analyze uncertainty-related language in FOMC discussions and show that higher uncertainty about the economic outlook predicts a more hawkish policy stance during meetings. Sharpe et al. (2023) and Aruoba and Drechsel (2025) find that a more optimistic tone in the Greenbook text predicts a higher federal funds rate. Finally, this paper relates to our previous work (e.g., Riboni and Ruge-Murcia (2010) and Riboni and Ruge-Murcia (2025)), but in that earlier work our focus was not on deliberation, but we used instead theoretical voting models to estimate the preferences of the median member and the chair on a hawk–dove scale, relying only on data from final policy decisions and the status quo. In contrast, this paper takes a completely different approach by analyzing deliberation directly through textual measures.

This paper is organized as follows. Section 2 describes the raw data employed in the analysis. Section 3 describes the text-based methods used to estimate the voiced policy stance and emotional tone of FOMC members. Section 4 presents an econometric model of deliberation within and across meetings, and discusses identification and estimation issues. Section 5 reports empirical estimates and robustness results, and studies the effect of transparency on deliberation. Section 6 examines the effects of deliberation and tone on interest rate decisions and dissents. Section 7 concludes.

2 Data

The raw data are transcripts from 358 FOMC meetings between March 1976 and December 2018 covering the chairmanships of Arthur Burns, William Miller, Paul Volcker, Alan Greenspan, Ben Bernanke, Janet Yellen, and Jerome Powell. A transcript records all statements by all participants in the meeting, identifying each speaker by name. A statement is a complete and uninterrupted intervention by a participant in a meeting and consists of one or more sentences. Transcripts are produced by Federal Reserve staff from the audio recording of the meetings and were originally to

serve only as input in the preparation of minutes. Their existence became known in October 1993 and the Federal Reserve subsequently agreed to release them to the public with a five-year lag. The total number of statements is 142,936, but we exclude statements by staff and bank vice-presidents (24,310). Our final sample consists of 118,626 statements by chairs, board governors, and bank presidents. The number of statements per meeting ranges from 111 on August 21, 2001, to 1,726 on February 3, 1981. The number of words per statement ranges from 1 to 4,731 (Mr. Fisher, President of the Dallas Fed in the meeting on May 16, 2000). Transcripts are available from the website of the Federal Reserve (www.federalreserve.gov).

Real-time data available to FOMC members at the time of the meeting were taken from the *Current Economic and Financial Conditions* or Greenbooks.³ Greenbooks are produced by Federal Reserve staff prior to each meeting and contain data, projections, and analysis on several economic variables. Greenbooks are circulated to all FOMC members roughly six days before the meeting. All Greenbooks produced for the meetings in our sample were obtained from the Real-Time Data Research Center at the Federal Reserve Bank of Philadelphia (www.philadelphiafed.org). In our empirical work we use the Greenbook variables that span the complete sample period, namely, inflation, output growth, unemployment, and housing starts.⁴ Inflation is the rate of growth of the GNP deflator (GDP deflator after 1992Q1) and output growth is the rate of growth of real GNP (real GDP after 1992Q1). Growth rates are annualized quarter-over-quarter in percentage points.

While most of our empirical analysis covers the period from March 1976 to December 2018, our analysis of interest rate decisions ends in 2008 with the financial crisis, when the federal funds rate reached its effective lower bound and quantitative easing became the primary policy instrument. The decisions on the federal funds target were taken from Appendices 4 and 5 in Chappell et al. (2005) for the period before February 1978 and from August 1987 to December 1996, from Thornton (2005) for the period from October 1982 to July 1987, and collected directly by the authors from transcripts and the *Record of Policy Actions* for the periods from March 1978 to October 1979 and from January 1997 to December 2008.

In some of our regressions we use lags of the federal funds rate target. However, because

³From June 2010 onward, a Tealbook combines the previous Greenbook and Bluebook. Since the former term is familiar in the literature, we will use it throughout this paper.

⁴The growth rate of the index of industrial production is also available for the complete sample period but we abstained from using this series because its correlation with output growth is high (0.86) and this raises the possibility of collinearity in the regressions below.

conference calls sometimes led to target changes and adjustments between meetings were common in the 1970s and 1980s, these lags do not necessarily correspond to values selected in previous FOMC meetings. Rudebusch (1995) and Thornton (2005) construct federal funds target series that include conference calls and inter-meeting policy adjustments. We use the series constructed by Rudebusch for the period from March 1976 to September 1979 and by Thornton for the period from September 1982 to December 1993. For the period after 1993, all inter-meeting adjustments were decided in conference calls and we use the transcripts from those calls to determine whether there was a change in the target and its magnitude if there was a change. By construction, this sample includes only meetings where the decision concerns the federal funds rate target and excludes meetings that concern other policy instruments (like non-borrowed reserves and asset purchases). This sub-sample consists of 254 face-to-face meetings between March 1976 and December 2008, excluding the meetings between October 1979 and October 1982 when the Federal Reserve pursued a policy of non-borrowed reserves targeting. The sample includes the meeting on October 6, 1979 because in that meeting, and in contrast to subsequent ones, the chair stated a precise, “ideal” value for the federal funds rate (see p. 55 of the transcript).

The data on dissents were taken from Thornton and Wheelock (2014). Their sample starts in 1936 and ends in 2013, but the series is updated and maintained by the Federal Reserve Bank of St. Louis and available from its website (www.fred.stlouisfed.org). Our sample is a subset of theirs because our time span is shorter (from 1976 to 2018) and we focus on regular meetings for which both transcripts and Greenbooks are produced.

In broad terms, FOMC meetings are segmented into two rounds: 1) an economic condition discussion (henceforth, the economic round), and 2) a policy discussion (henceforth, the policy round), which are followed by the chair’s proposal and voting. The former round involves the presentation of economic conditions and forecasts by staff and of the views on the economic outlook by FOMC members. The latter round involves the discussion of policy alternatives and preferences by FOMC members. The transition between the two rounds is usually signaled by the beginning of the monetary policy briefing, which is often marked by the chair’s introduction or by a detailed presentation of policy alternatives. In our analysis, all discussions preceding this signal are classified into the economic round. Discussions on long-term monetary growth targets, and infrequent miscellaneous discussions—such as those on legal matters, simulation techniques, or specific studies—that usually

take place before the formal start of the meeting, are classified into the economic round as well. However, these discussions are infrequent and the round they are assigned to is unlikely to impact our results. Lastly, a small number of meetings exhibit intermingled discussions, in which case a coffee break serves as the separating marker between rounds.

3 Quantifying Speech

We use text mining and artificial intelligence (AI) techniques to characterize quantitatively the speech of FOMC members along several dimensions. The input are the 118,626 statements by chairs (46,515), board governors (28,630), and bank presidents regardless of whether they are voting members or not (43,381). The statistics described below were computed at the statement level, i.e., for each intervention by each meeting participant. The figures were then aggregated to obtain a measure for each meeting for each of three FOMC membership classes—chairs, board governors, and bank presidents—using as weight the ratio of the number of words in the sentence to the total number of words pronounced by the members of that class in the meeting.

3.1 Voiced Policy Stance

The first set of variables are policy-stance scores that measure a member’s voiced support for tighter or looser monetary policy. The first measure is a rule-based measure based on Apel and Grimaldi (2014) and Malmendier et al. (2021). The measure is based on identifying pairs with a goal keyword (e.g., "inflation") and a word indicative of either tighter policy (e.g., "increase") or looser policy (e.g., "decrease") in a two-word window around the goal keyword.⁵ Then, for a given statement, the tighter- and looser-policy stances are:

$$\begin{aligned} \text{tighter stance rule } (i, t) &= \frac{\#tighter\ pairs(i, t)}{\#tighter\ pairs(i, t) + \#looser\ pairs(i, t)}, \\ \text{looser stance rule } (i, t) &= \frac{\#looser\ pairs(i, t)}{\#tighter\ pairs(i, t) + \#looser\ pairs(i, t)}, \end{aligned} \tag{1}$$

⁵The goal keywords are "inflation", "growth", "price", "prices", "wages", "employment", "unemployment", "GDP", "output", "activity", and "job." The tight-policy keywords are "increase", "higher", "raise", "rise", "tighten", "strong", "fast", "gain", "hike", "growth", "accelerate", "expand", "surge", "boom", "strengthen", "up", "lift", and "climb." The looser-policy keywords are "cut", "decrease", "down", "ease", "fall", "reduce", "deflate", "weak", "weaken", "slow", "loss", "slowdown", "decelerate", "mitigate", "contract", "slice", "shave", "trim", and "drop."

where (i, t) refers to statement i in meeting t . By construction, (1) sets the stance to zero if no pairs are found in statement (i, t) . As Apel and Grimaldi (2014), we abstain from considering negations. Section 6 examines the robustness of our results to an alternative calculation of policy stances.

Apel and Grimaldi (2014) and Malmendier et al. (2021) interpret their measure as an indicator of the speaker’s preferences (hawkish *versus* dovish). We limit the interpretation of our measure to that of a *voiced* preference, leading to, or consistent with, a policy action (tighter *versus* looser). In our econometric model of deliberation, voiced preferences depend on information available at the beginning of the meeting—notably variables from the Greenbook—and on information exchanged with other members through deliberation.⁶ Our interpretation addresses the concern that, for example, if inflation is high, members may mention this in their statements and support a tighter policy regardless of whether they are intrinsically hawkish or dovish.

The second measure employs a language model based on *RoBERTa-large* (Liu et al. (2019)) and trained using transfer learning to obtain predictive labels (tighter policy, looser policy, or neutral) for each sentence in statement. The label takes the discrete values zero or one. *RoBERTa* is a transformer model—that is, a neural network that learns context and meaning by tracking relationships in sequential words in a sentence—pretrained in a self-supervised manner on a large corpus of English data. Transfer learning is a machine learning technique where a model pretrained on one task is fine-tuned for a related task, building on the knowledge from the first task to enhance performance on the new one. Shah et al. (2023) apply this technique to FOMC speeches, meeting minutes, and press conference transcripts to construct a measure of monetary policy stance and examine its relation to inflation and Treasury bill yields. The stance at the statement level is computed as,

$$\begin{aligned} \text{tighter stance language } (i, t) &= \frac{\text{tighter label } (i, t)}{\text{tighter label } (i, t) + \text{looser label } (i, t)}, \\ \text{looser stance language } (i, t) &= \frac{\text{looser label } (i, t)}{\text{tighter label } (i, t) + \text{looser label } (i, t)}. \end{aligned} \tag{2}$$

By construction, (2) sets neutral statements, i.e., those without a tighter or looser label, to zero.

⁶In principle, voiced preferences may also implicitly depend on the speaker’s preferences, for instance, as in Riboni and Ruge-Murcia (2025) where the relative hawkishness/dovishness of a member is captured by a fixed intercept in the reaction function. However, given the level of aggregation in our empirical analysis, we abstained from pursuing this implication further.

For our empirical analysis, we combine the policy stance of the three broad classes of FOMC membership, namely chairs, board governors, and bank presidents, into a single measure for each.⁷ This approach is based on 1) institutional information that indicates that the chair plays a unique role within the FOMC, and 2) previous research that shows that voting and dissent patterns between board governors and bank presidents are different. Differences in voting behavior between governors and bank presidents were first noted by Yohe (1966) and Canterbury (1967). Subsequent research on dissents finds that governors tend to prefer more expansionary policies than bank presidents (see Belden (1989), Gildea (1990), and Riboni and Ruge-Murcia (2014)).⁸ Eijffinger et al. (2015) identify the policy preferences of FOMC members based on their statements during the policy round and find that the ideal point of the median board governor is generally more dovish than that of the median bank president. Bordo and Istrefi (2023) find that board governors are perceived by the media to be more dovish than bank presidents. In sum, this research suggests that distinguishing between board governors and bank presidents is a natural partition of the FOMC membership.

We treat the stances for tighter and looser monetary policy separately for several reasons. First, in basically all meetings and rounds, members voice stances for both policies. The proportions of meetings/rounds where this is the case are reported in Table 1 for both measures and range from 66% to 100%. These results suggest that regardless of what the final vote may indicate, there is seldom unanimity among FOMC members as to the best policy to follow and deliberation is, therefore, critical for reaching a decision. Second, voiced tighter and looser policy stances involve distinct communication strategies, often using markedly different vocabularies, among other reasons because they are associated with different risks and constraints (e.g., the zero lower-bound in the case of a looser policy). In other words, tighter and looser policy stances are not mirror images of each other. Third, our aggregation across membership classes implies that individuals in the same class (say, two bank presidents) may voice different stances in a given meeting. Thus, treating stances for tighter and looser policy separately allows us to explicitly consider heterogeneity in preferred policies. Finally, by treating both stances separately, we can study their interaction

⁷In preliminary work, we considered performing our econometric analysis at the highly disaggregated individual level, but found the data to be too noisy to allow meaningful conclusions. For the same reason, we abstained from studying subsamples for individual chairs as each features a relatively small number of observations: 24 for Burns, 15 for Miller, 67 for Volcker, 149 for Greenspan, 64 for Bernanke, 32 for Yellen, and 7 for Powell.

⁸In our sample, board governors and bank presidents cast a total of 88 and 147 dissenting votes, respectively. In the case of board governors, 44% of their dissenting votes were for a tighter policy, while in the case of bank presidents, the figure is 85%.

during deliberation and across meetings.

Figure 1 shows that although measures (1) and (2) are constructed using different methodologies, they are consistent in their estimation of the voiced policy stance for all members and meetings in the sample. Put differently, both (1) and (2) agree in their assessment of the policy stance with correlations of 0.65 and 0.54 for the tighter and looser stances, respectively. If the correlations were exactly 1, then either of the two measures (rule- or language-based) could be used in empirical work without any loss of information. Instead, since the correlations are high but not identical to 1, there is a gain from combining both measures into one for each stance (tighter and looser). In particular, our final estimate of tighter (T) and looser (L) policy stances is the first principal component of their respective two measures in (1) and (2). This component is the linear combination of the measures that accounts for largest proportion of their variance, with coefficients given by the elements of the eigenvector associated with the largest eigenvalue of the correlation matrix. This eigenvector is unique up to a linear transformation, and we normalize its sign so that the interpretation of the principal component is consistent with that of the underlying textual series. That is, if a larger value of the textual series means a stronger stance for tighter policy, then the same is true for their first principal component.⁹ Before running the principal component analysis (PCA), we standardized the series and carried out Bartlett tests of the hypothesis that their correlation matrix is the identity matrix. Test results are reported in Table 2 and show that the hypothesis can be rejected at standard significance levels in all cases. This conclusion justifies our use of PCA to summarize the information from both series in a single measure. Finally, we control for long-term trends in language patterns by regressing the principal component on a constant and a quadratic time trend, with the resulting residuals becoming our final measure of policy stance.¹⁰ We do the same for our measures of emotion and collaboration below.

Our final measures of voiced policy stance are plotted in Figure 2 for chairs, board governors, and bank presidents. Their cross correlations are reported in Table 3. The correlations of stances for the same policy (tighter or looser) are positive across member classes and range from 0.30 to

⁹Note that in contrast to, e.g., Handlan (2022), who re-scales the first principal component so that a one-unit increase corresponds to a 100 basis point increase in the change of a treasury yield, in our application there is no benefit from normalizing the size of the eigenvector because the underlying series are unit-free indices.

¹⁰Trends in the usage of the English language are well documented in the linguistics literature. For instance, see Bauer (1994) who examines changes in lexicon (vocabulary), grammar, and sounds in English throughout the twentieth century.

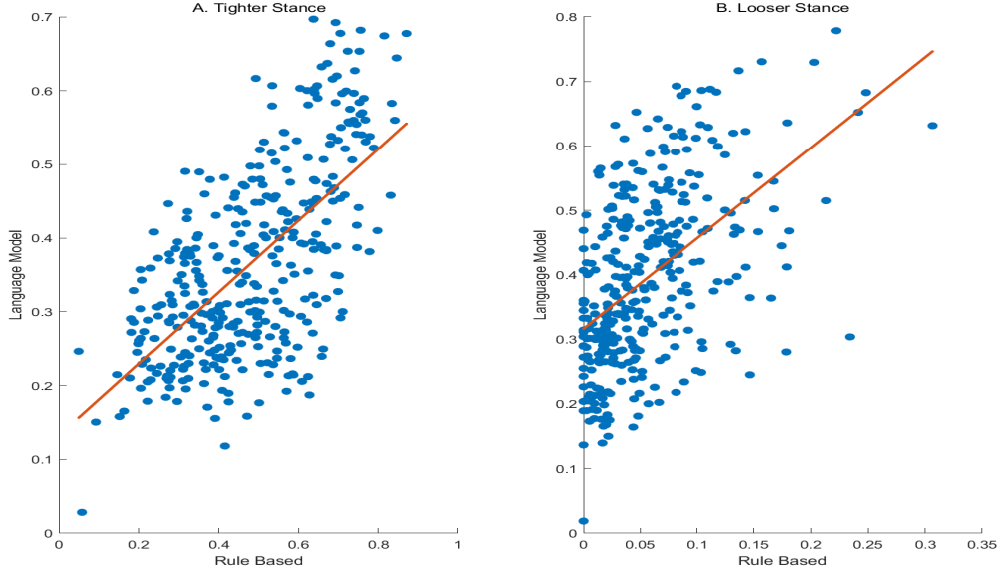


Figure 1: Relation Between Rule-based and Language Model Estimates of Policy Stance

0.75, but they are well below 1. The correlations of stances for a different policy (tighter *vs.* looser) are negative across member classes and range from -0.52 to -0.12 , but they are well above -1 . These results mean that our two measures of stance (tighter and looser) carry different information and, in addition to the reasons above, motivate our strategy of treating them separately. Figure 2 also illustrates the inherent noisiness of the textual measures that explain our decision to aggregate them at the level of member classes and for substantial parts of, or the whole, meeting.

3.2 Emotions

The second set of variables are measures of emotion. The measures are based on a seven-emotion classification computed following Hartmann (2022). Hartmann fine-tuned the transformer model *DistilRoBERTa-base* using manual labels from Twitter, Reddit, student self-reports, and television dialogues. The seven emotions are anger, disgust, fear, happiness, neutral, sadness, and surprise. As in previous literature, a distinction is made between negative and positive emotions. Gorodnichenko et al. (2023) use a deep learning model to detect emotions from the voice tone in press conferences that follow FOMC meetings and focus on negative (anger and sadness) and positive (happiness and pleasantly surprised) emotions. Curti and Kazinnik (2023) and Alexopoulos et al. (2024) use facial

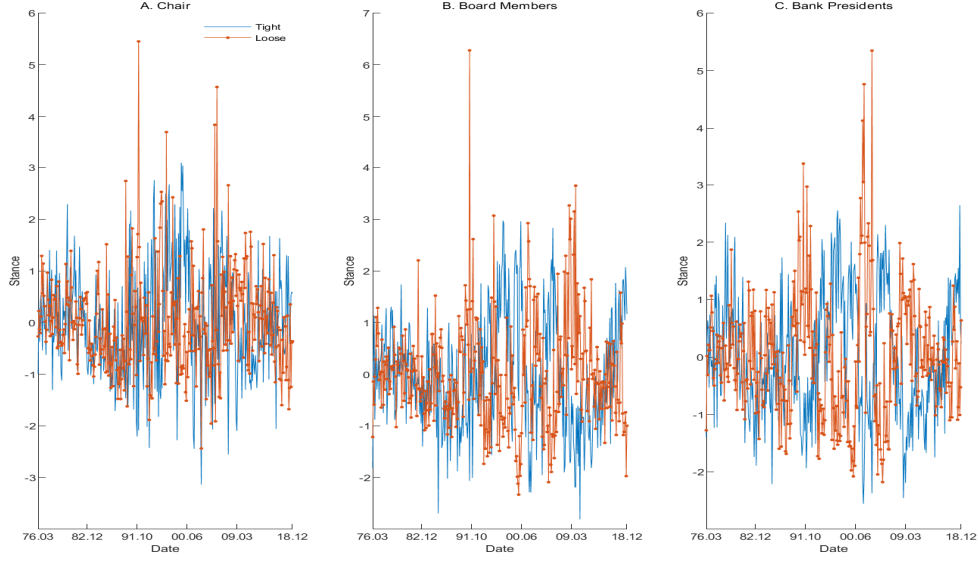


Figure 2: Voiced Policy Stances

recognition technology to infer emotions from press conferences and congressional testimonies by Federal Reserve chairs, respectively. Curti and Kazinnik (2023) focus on negative emotions and control for tone and policy stance. Alexopoulos et al. (2024) average the four negative emotions and control for vocal pitch and textual emotion. All these papers use a narrow identification window to estimate the effects of emotions on the financial markets.¹¹

An emotion measure at the statement level is constructed as

$$emotion(i, t) = \frac{1}{Q} \sum_{j=1}^Q \mathcal{S}(sentence_j = emotion), \quad (3)$$

where j denotes the sentence in the statement, Q is the number of sentences in the statement, and $\mathcal{S}(\cdot)$ is an indicator function that takes value 1 if the sentence features the given emotion and 0 otherwise. Our final estimate of negative emotion, N , is the first principal component of anger, disgust, fear and sadness. See Table 1 for the results of Bartlett tests that justify the use of PCA.¹² Our estimate of positive emotion, P , is based only on happiness and standardized to

¹¹See also Mathur et al. (2022) who use videos from monetary policy calls for six central banks to examine whether visual, vocal, and verbal behavioral cues are helpful in forecasting financial asset prices and volatility in days following the calls.

¹²Notice, however, that the p -value of the hypothesis concerning the correlation matrix of the measures of negative

be consistent with how we treat other variables in the PCA.¹³ We stress that positive emotion is not merely negative emotion with an opposite sign, but that it measures a distinct tone in the deliberation. For instance, while their correlation is negative (about -0.20), as one would expect, it is much higher than -1 , as would be the case if they were simply alternative measures of the same object. Moreover, it is entirely possible for positive and negative emotions to increase or decrease simultaneously in the data, reflecting shifts in the overall tone or intensity of deliberation rather than a simple trade-off between the two emotions.

3.3 Collaboration

The third set of variables are displays of personality traits helpful in the deliberative process. Wang and Sun (2024) use the *PANDORA* dataset and a fine-tuned version of *RoBERTa-base* to develop models that predict five key traits: extroversion, neuroticism, conscientiousness, agreeableness, and openness. These five traits constitute the well-known “Big Five” personality model, which is widely employed in psychological and behavioral research, but also studied in economics (e.g., see Heckman et al. (2021)). The *PANDORA* dataset includes extensive personality labeling of Reddit comments from over 10,000 users, labeled with three personality models (Enneagram, Big Five, and MBTI) and demographic data (such as age, gender, and location). We focus on agreeableness (measuring helpfulness and willingness to cooperate) and openness (measuring creativity, curiosity, and willingness to entertain new ideas) as they are especially relevant for collective decision-making. Sentences are individually labeled, but in contrast to the emotion labels, this label is a continuous variable rather than a dummy variable, and a sentence can feature more than one trait. A measure of the trait at the statement level is constructed as

$$trait(i, t) = \frac{1}{Q} \sum_{j=1}^Q \mathcal{T}(trait_j), \quad (4)$$

where $\mathcal{T}(\cdot)$ is the value of the label for sentence j . The estimate of collaboration, C , is the first principal component of agreeableness and openness. See Table 1 for the results of Bartlett tests

emotion expressed by the chair for the whole meetings is 0.105. Thus, the hypothesis would not be rejected at the 10% level, though this result is clearly marginal.

¹³In preliminary work we also treated surprise as a positive emotion, but found inconclusive results because we cannot distinguish between good and bad surprises in the data.

that justify the use of PCA. Our final measures of tone (negative emotion, positive emotion, and collaboration) used in the empirical analysis below are plotted in Figure 3. Since our goal is to characterize the overall tone of a meeting, we do not distinguish tone across membership classes.

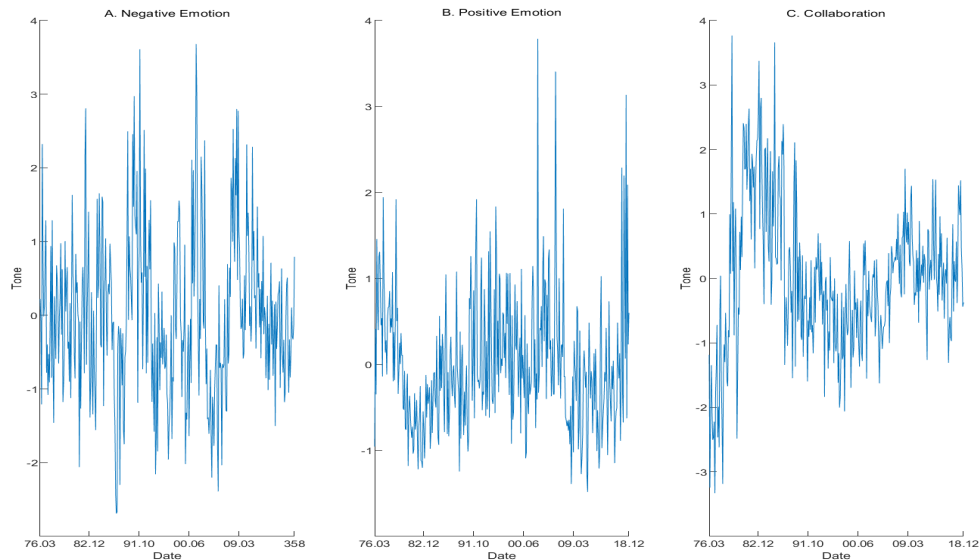


Figure 3: Tone in FOMC Meetings

4 An Econometric Model of Deliberation

Equipped with quantitative measures of voiced policy stance and tone, we construct and estimate a model of sequential deliberation. In broad terms, our econometric model is inspired by the theoretical literature on deliberation (e.g., Coughlan (2000) and Austen-Smith and Feddersen (2006)) where members are allowed to engage in pre-vote communication prior to voting. The model describes the meeting as a sequence of deliberation rounds and can be represented in reduced-form as vector auto-regression (VAR) with a terminal observation at the end of the meeting. A generalization of the model is then used to examine longer-term effects of deliberation across meetings.

4.1 Model

Consider a monetary committee composed of M members and tasked with selecting the level of a policy instrument (e.g., the key nominal interest rate).¹⁴ The committee arrives at its decision in a deliberative meeting. Before the meeting, the information, Ω , is simultaneously revealed to all members.¹⁵ The meeting is divided into two or more deliberation rounds $r = 1, 2, \dots, R$, and the policy decision made at the end of the last round.¹⁶

In any given round, r , members deliberate and each member expresses a policy stance and a tone that depends on the initial information and the stance and tone of other members. The deliberation process is represented by means of the recursive functions,

$$\begin{aligned}\mathbf{S}_r &= \mathcal{F}(\Omega, \mathbf{S}_r, \mathbf{X}_r) + \zeta_r, \\ \mathbf{X}_r &= \mathcal{G}(\Omega, \mathbf{X}_r, \mathbf{S}_r) + \xi_r,\end{aligned}\tag{5}$$

where $\mathbf{S}_r = [\mathbf{T}_r \ \mathbf{L}_r]$ is a $2M \times 1$ vector with the policy stances of all members, $\mathbf{X}_r = [\mathbf{N}_r \ \mathbf{P}_r \ \mathbf{C}_r]$ is a $3M \times 1$ vector with the tone of all members, and ζ_t and ξ_r are respectively $2M \times 1$ and $3M \times 1$ vectors of serially correlated disturbances with mean zero. Note that $\mathbf{S}_r$ is obtained by stacking the $M \times 1$ vectors, $\mathbf{T}_r$ and $\mathbf{L}_r$, that respectively contain the tighter and looser policy stances of the M members, and $\mathbf{X}_r$ is obtained by stacking the $M \times 1$ vectors, $\mathbf{N}_r$, $\mathbf{P}_r$, and $\mathbf{C}_r$, that contain the tone characteristics of their interventions in round r .

Three important features of (5) are the following. First, the information in Ω is predetermined, common to all members, and fixed throughout the meeting. Second, the parameters of $\mathcal{F}(\cdot)$ and $\mathcal{G}(\cdot)$ may be member-specific, meaning that members can weight differently the information contained in Ω , and the stance and tone of other members. Finally, both features of the discussion—policy stance and tone—in round r are jointly and simultaneously determined in the deliberation process.

¹⁴For the estimation of the model using the FOMC data constructed in Section 3, we assume $M = 3$ corresponding to the chair, governors, and bank presidents. For a justification of this partition of the FOMC membership, see Section 3.1.

¹⁵In the case of the FOMC, empirical analogues of Ω are 1) the economic data contained in the Greenbooks circulated to all members before each meeting, and 2) the staff presentations by the manager of the System Open Market Account at the Federal Reserve Bank of New York and by the directors of the Division of Research and Statistics and the Division of International Finance at the beginning of each meeting.

¹⁶In the case of the FOMC, we assume $R = 2$ with $r = 1$ being the economic round and $r = 2$ being the policy round.

In what follows, it is convenient to assume that $\mathcal{F}(\cdot)$ and $\mathcal{G}(\cdot)$ are linear functions as in,

$$\begin{aligned}\mathbf{S}_r &= \alpha_1 + \beta_1 \Omega + \lambda_1 \mathbf{S}_r + \lambda_2 \mathbf{X}_r + \zeta_r, \\ \mathbf{X}_r &= \alpha_2 + \beta_2 \Omega + \delta_1 \mathbf{X}_r + \delta_2 \mathbf{S}_r + \xi_r.\end{aligned}\tag{6}$$

where α_1 and α_2 are respectively $2M \times 1$ and $3M \times 1$ vectors of constant intercepts; β_1 and β_2 are respectively $2M \times K$ and $3M \times K$ matrices of coefficients, with K the number of variables in Ω ; and λ_1 , λ_2 , δ_1 , and δ_2 are conformable matrices of coefficients. Assume that ζ_r and ξ_r jointly follow the AR(1) process,

$$\begin{bmatrix} \zeta_r \\ \xi_r \end{bmatrix} = \Gamma \begin{bmatrix} \zeta_{r-1} \\ \xi_{r-1} \end{bmatrix} + \begin{bmatrix} u_r \\ v_r \end{bmatrix},\tag{7}$$

where

$$\Gamma = \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix},\tag{8}$$

and u_r and v_r are mutually and serially uncorrelated innovations with diagonal variance-covariance matrix Σ . Since the variance of the innovations to the policy stance and tone need not be the same across members, we assume that Σ is heteroskedastic. The sub-matrices γ_{11} , γ_{12} , γ_{21} , and γ_{22} take values such that all eigenvalues of Γ are inside the unit circle and the process of $[\zeta_r \ \xi_r]'$ is stationary.

Consider the first deliberation round, $r = 1$. Solving for the policy stance and tone,

$$\begin{bmatrix} \mathbf{S}_r \\ \mathbf{X}_r \end{bmatrix} = \Lambda^{-1} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} + \Lambda^{-1} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \Omega + \Lambda^{-1} \begin{bmatrix} \zeta_r \\ \xi_r \end{bmatrix},\tag{9}$$

where

$$\Lambda = \begin{bmatrix} I - \lambda_1 & -\lambda_2 \\ -\delta_2 & I - \delta_1 \end{bmatrix},\tag{10}$$

I is an identity matrix of appropriate size, and we assume that the disturbances $[\zeta_r \ \xi_r]'$ for this first round are drawn from their unconditional distribution, which has mean zero and variance given by $vec(\Psi) = (I - \Gamma \otimes \Gamma)^{-1} vec(\Sigma)$. Consider now the subsequent deliberation rounds. Quasi-differencing (6) and solving for the policy stance and tone,

$$\begin{bmatrix} \mathbf{S}_r \\ \mathbf{X}_r \end{bmatrix} = \Lambda^{-1}(I - \Gamma) \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} + \Lambda^{-1}\Gamma\Lambda \begin{bmatrix} \mathbf{S}_{r-1} \\ \mathbf{X}_{r-1} \end{bmatrix} + \Lambda^{-1}(I - \Gamma) \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \Omega + \Lambda^{-1} \begin{bmatrix} u_r \\ v_r \end{bmatrix},\tag{11}$$

for $r = 2, 3, \dots, R$, where we have used the fact that the information, Ω , is the same in all rounds of a meeting. Note that the policy stance and tone jointly follow a vector auto-regression of order 1 throughout the rounds of the same meeting, with the exogenous regressor, Ω , and a terminal observation in the last round $r = R$.

The final policy decision by the committee is a function of economic data (inflation, output growth, etc.), policy stances and tone voiced by all members during the meeting, and past policy decision (e.g., due to policy smoothing):

$$\begin{aligned} I_t &= \mathcal{I}(I_{t-1}, \Omega_t, \mathbf{S}_t, \mathbf{X}_t) + \mu_t, \\ &= \phi_1 + \phi_2 I_{t-1} + \phi_3 \Omega_t + \phi_4 \mathbf{S}_t + \phi_5 \mathbf{X}_t + \mu_t, \end{aligned} \tag{12}$$

where t denotes the date of the meeting, μ_t is a disturbance term, and the time subscript makes clear that the economic data, stance, and tone will be different in different meetings. To simplify the notation, (12) contains only one lag of the policy decision, but in the empirical work below we incorporate a sufficient number of lagged policy decisions so that μ_t is serially uncorrelated.¹⁷

After the policy decision is agreed by the committee, one or more members may dissent. We assume

$$D_t = \mathcal{D}(\Omega_t, \mathbf{S}_{R,t}, \mathbf{X}_{R,t}) + \epsilon_t, \tag{13}$$

where ϵ_t is a disturbance term and $\mathcal{D}(\cdot)$ is a nonlinear function of the economic data and the policy stances and tone voiced by all members in the last round of the meeting (i.e., the policy round in the case of the FOMC).¹⁸ In the empirical work below, we will take a stance on the precise form of $\mathcal{D}(\cdot)$ to represent the fact that whether we observe a dissent or not is a binary variable and the dissent rate (another variable of interest) is bounded below by zero.

¹⁷Coibion and Gorodnichenko (2012) report evidence that interest rate smoothing by the Federal Reserve takes the form of an autoregressive process of order higher than 1 and the autoregressive coefficients of the disturbance term (our μ_t) are insignificantly different from zero or negative.

¹⁸We adopt this specification because it is natural to assume that dissent in the FOMC arises directly from disagreement about policy as voiced in the policy round. Chappell et al. (2005), Meade (2005) and Eijffinger et al. (2015) note that it is in this round where meeting participants first express their policy preferences explicitly.

4.2 Reduced Form, Identification, and Estimation

The structural model of deliberation in (9) and (11) has the reduced-form,

$$\begin{bmatrix} \mathbf{S}_r \\ \mathbf{X}_r \end{bmatrix} = \mathbf{A} + \mathbf{B}\Omega + \mathbf{U}_r, \quad (14)$$

for $r = 1$, and

$$\begin{bmatrix} \mathbf{S}_r \\ \mathbf{X}_r \end{bmatrix} = \mathbf{C} + \mathbf{D} \begin{bmatrix} \mathbf{S}_{r-1} \\ \mathbf{X}_{r-1} \end{bmatrix} + \mathbf{H}\Omega + \mathbf{V}_r, \quad (15)$$

for $r = 2, 3, \dots, R$, where the disturbances in $\mathbf{U}_r$ and $\mathbf{V}_r$ are serially uncorrelated.¹⁹

The structural model implies that the policy stance and tone in the first round of the meeting are determined by the interaction of policy stance and tone among committee members, with the fundamental driver being the information contained in Ω . However, the reduced-form model above makes clear that the contemporaneous effect of stance on tone and vice-versa (that is, λ_2 and δ_2) is not identified. In contrast, the total effect of information on policy stance and tone is identified in (14). The total effect includes the direct effect (β_1 and β_2) and the effect from deliberation (captured by Λ in (9)), both of which are subsumed in the matrix $\mathbf{B}$ in (14).

The structural model also implies that the current stance and tone have predictive power over their values in subsequent rounds, beyond the information contained in Ω . This prediction is testable by means of t and F tests of the appropriate coefficients of $\mathbf{D}$ in (15). The total effect of information (Ω) on the policy stance and tone in rounds $r = 2, 3, \dots, R$ is testable as well.

Estimation of the reduced form of the model is carried out by Ordinary Least Squares (OLS) line by line in (14) and (15). Standard econometric results (e.g., Hamilton (1994) Chap. 11) show that the coefficients found by these OLS regressions are also the Maximum Likelihood (ML) estimates under the additional assumption that disturbances are normally distributed. Since the variance-covariance matrix of the innovations in the structural model, Σ , is heteroskedastic, the disturbances $\mathbf{U}_r$ and $\mathbf{V}_r$ in the reduced-form are heteroskedastic as well. For this reason, we compute and report Huber–White heteroskedasticity-consistent standard errors of the OLS coefficients.

¹⁹This follows from the assumptions that the initial draw of ζ_t and ξ_r in $r = 1$ comes from their unconditional distribution and do not depend on the past, and that the innovations to the AR process in (7) are serially uncorrelated.

4.3 A Generalization

Consider a generalization of the model where deliberation also takes place across meetings, in addition to rounds within a meeting. That is, the voiced policy stances and tone by members in one meeting may affect those in the subsequent meeting. Evidence of effects across meetings is based on earlier work on dissents and on anecdotal accounts from previous FOMC members. Regarding the former, Riboni and Ruge-Murcia (2014) report empirical evidence from the Monetary Policy Committee of the Bank of England that a current dissent by a committee member helps forecast future votes of other members. Regarding the latter, several former members state that their interventions were meant to influence other members' views more in future meetings than in the current one (see Meyer (2006) and Schonhardt-Bailey and Bailey (2013)). For example, A. Blinder and L. Lindsey respectively note that “[Y]ou would make a point, maybe forcefully if necessary, at one meeting in the hope that it would sink in and by the next meeting . . . or the meeting after that, you are starting to persuade people” and “I viewed the discussion at the meetings not necessarily as useful for influencing the vote of that day but as influencing the way the thinking was going to go, down the [road], in future meetings”.²⁰ In the same spirit, Meyer (2006) writes “I always listened to the discussion intently, because it could change my mind, even if it could not change my vote at that meeting. Similarly, while in my remarks to my colleagues it sounded as if I were addressing today’s concerns and today’s policy decisions, in reality I was often positioning myself, and my peers, for the next meeting” (see p. 53).

The equation corresponding to (11) in the case of deliberation across meeting is obtained by quasi-differencing (6), solving for the policy stance and tone, and noting that the information set is different across meetings. Thus,

$$\begin{bmatrix} \mathbf{S}_t \\ \mathbf{X}_t \end{bmatrix} = \Lambda^{-1}(I - \Gamma) \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} + \Lambda^{-1}\Gamma\Lambda \begin{bmatrix} \mathbf{S}_{t-1} \\ \mathbf{X}_{t-1} \end{bmatrix} + \Lambda^{-1} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \Omega_t - \Lambda^{-1}\Gamma \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \Omega_{t-1} + \Lambda^{-1} \begin{bmatrix} u_t \\ v_t \end{bmatrix}, \quad (16)$$

with reduced-form

$$\begin{bmatrix} \mathbf{S}_t \\ \mathbf{X}_t \end{bmatrix} = \mathbf{C} + \mathbf{D} \begin{bmatrix} \mathbf{S}_{t-1} \\ \mathbf{X}_{t-1} \end{bmatrix} + \mathbf{H}\Omega_t + \mathbf{K}\Omega_{t-1} + \mathbf{W}_t, \quad (17)$$

²⁰These two, and similar statements, can be found in Section 5.3.1 in Schonhardt-Bailey and Bailey (2013).

where $\mathbf{W}_t$ is serially uncorrelated and heteroskedastic. In this case, the policy stance and tone jointly follow a VAR of order 1 with exogenous regressors, Ω_t and Ω_{t-1} . Compared with the within-meeting model of deliberation, this VAR does not have a terminal observation. In line with the anecdotal evidence above, the VAR implies that the voiced policy stance and tone in one meeting help forecast, or Grange-cause, the policy stance and tone in the subsequent meeting, after controlling for the economic conditions in both meetings. This prediction is testable and the corresponding results are reported below. Estimation is carried out by Ordinary Least Squares (OLS) line by line with Huber–White heteroskedasticity-consistent estimates of standard errors.

5 Estimates

This section reports estimates from the deliberation model within meetings and across meetings, and robustness results.

5.1 Economic Round

Estimates of the deliberation parameters in the first round of FOMC meetings (i.e., the economic round) are reported in Table 4, with t -statistics reported in parenthesis. Panels A and B answer the questions of whether and how the voiced policy stance of the chair, board governors, and bank presidents reflect current economic conditions as reported in the Greenbooks.

First, note that the chair appears to be relatively restrained in the first round of the meeting and his/her policy stance cannot be fully predicted on the basis of observable economic data. None of the coefficients of the explanatory variables of his/her tighter-policy stance is statistically different from zero at the 5% level and the R^2 of the regression is very low: 0.01. The chair’s looser-policy stance is somewhat more predictable, with the unemployment coefficient significant at the 5% level and $R^2 = 0.11$. The positive estimate in Table 4 means that an increase in unemployment predicts a looser voiced policy stance by the chair in the economic round of the FOMC meeting. None of the other coefficients is statistically significant. These results may be partly explained by the fact that some chairs tend to speak relatively little during this round (see Hansen et al. (2018)) and, more generally, by the heterogeneity as to when chairs presented their views on the economy. For instance, Greenspan presented his views at the beginning of the policy round and most of

his interventions in the economic round were relatively neutral. In contrast, Bernanke and Yellen presented their views on the economy at the end of the economic round.

Second, in contrast to the chair, the tighter policy stance of board governors and bank presidents during the economic round can be predicted on the basis of data contained in the Greenbooks. Since the purpose of the economic round is precisely for both board governors and bank presidents to provide their perspectives on the current state of the economy, it is reassuring (for the validity of our measures) that their speeches do reflect economic conditions. In particular, an increase in output growth and a decrease in unemployment lead both members to voice a tighter monetary policy stance. The R^2 of the regressions are 0.20 and 0.30, respectively. Likewise, a decrease in output growth and an increase in unemployment lead both members to voice a preference for a looser policy. Overall, inflation and housing starts are not as helpful in predicting their policy stances in the first round of the meeting, but the coefficient of inflation in Panel B for board governors and bank presidents is significant at the 10% and 15% levels, respectively. Its negative sign means that a decrease in inflation leads both members to voice a stance for looser policy. These results imply an asymmetry in that an increase in inflation does not predict a tighter policy stance by board governors and bank presidents in the first round of the meeting, but a decrease in inflation may predict a looser policy stance. This asymmetry may be driven by the zero-lower bound on the nominal interest rate. For instance, in periods where the bound is binding, or close to binding, the FOMC may delay liftoff until it is highly confident that inflation is returning to target because of the risk management issues discussed, e.g., in Evans et al. (2015). Relatedly, it may be due to the fact that, while lower output growth is generally undesirable, higher inflation may not be necessarily harmful when it is below target. For example, the asymmetric loss functions reported in Figure 3 of Shapiro and Wilson (2021) show that when inflation is below 1.5%, FOMC members become more negative as inflation decreases.

Panel C shows that economic conditions have an effect on the tone during the first round of FOMC deliberation. A decrease in the rate of inflation and an increase in house starts increase positive emotion among committee members, while a decrease in output growth or an increase in unemployment increases negative emotion. In line with Shapiro and Wilson (2021)), we also find that an increase in inflation decreases negative emotion. In order to shed some light on these results, we plot in Panel A of Figure 4 the relation between output growth and negative emotion in

the first round of the meetings, with filled red dots denoting meetings that took place in a month deemed by the NBER to be in a contraction. Panel B plots the relation between inflation and negative emotion, with filled red dots denoting the meetings that took place in the months during Volcker’s chairmanship until December 1982, which is generally regarded as the year when the high U.S. inflation episode came to an end. Panel A clearly shows that negative emotion among FOMC members is generally high during contractions and these observations drive the finding of a negative growth coefficient (-0.133) in Table 4. Panel B shows that the finding of a negative inflation coefficient (-0.061) in Table 4 is driven by a below-average negative emotion during the high-inflation in the early 1980s. While this result may appear surprising, it is consistent with our finding below of increased collaboration among FOMC members during this period.

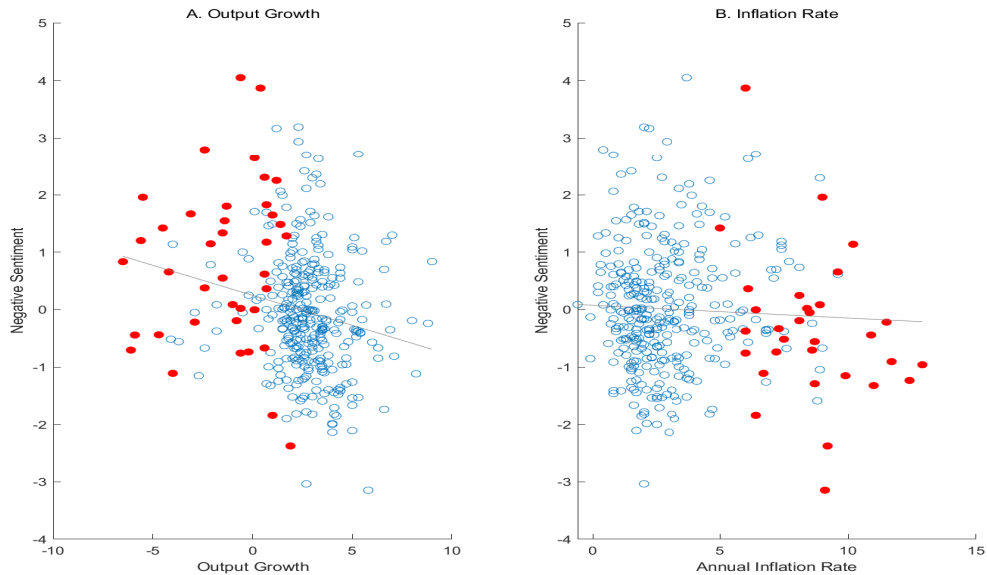


Figure 4: Relation Between Negative Emotions and Economic Conditions

Economic conditions also affect the members’ willingness to collaborate, with a decrease in output growth and increase in inflation leading to more collaboration among committee members. Panel A of Figure 5 plots the relation between output growth and collaboration in the first round of the meeting, with filled red dots denoting meetings that took place during a contraction. Panel B plots the relation between inflation and collaboration, with filled red dots denoting the meetings that took place during Volcker’s chairmanship until December 1982. From this figure, we conclude

that the regression results in Table 4 are driven by the fact that FOMC members appear to be more agreeable and open to collaboration during periods of economic contraction and high inflation.

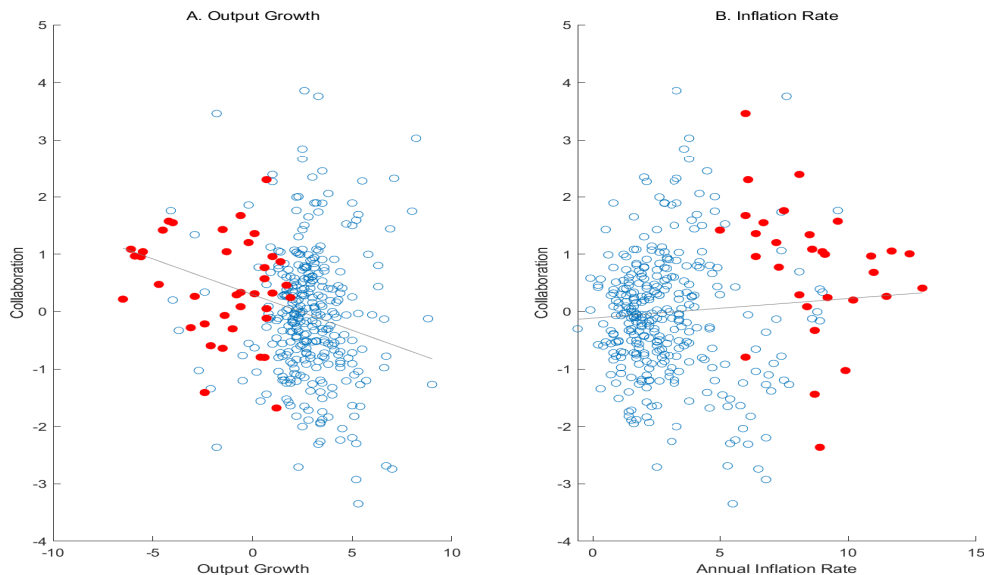


Figure 5: Relation Between Collaboration and Economic Conditions

5.2 Policy Round

Estimates of the deliberation parameters in the second round of FOMC meetings (i.e., the policy round) are reported in Table 5. This table answers the question of whether the voiced policy stance by the chair, board governors, and bank presidents, and the tone of the meeting in the second part of the meeting can be explained statistically by the stances and tone in the first part of the meeting, controlling for current economic conditions.

An important result is that voiced policy stances in the economic round have predictive power for the stance in the policy round within and across membership. A tighter policy stance by board governors in the economic round predicts a tighter policy stance by all members in the policy round. To see this, note that the coefficients of the regressor “Lag tight board” in the first three columns of Table 5 are all positive and statistically significant, two at the 5% level and one at the 10% level. Conversely, a looser policy stance by bank presidents in the economic round predicts a less tight policy stance by all members in the policy round. Note that the coefficients of the regressor

”Lag loose bank” in the first three columns are all negative and statistically significant, two at the 5% level and one at the 10%. Previous literature based on expressed dissents (e.g., Belden (1989)) and news coverage (Istrefi (2018) and Bordo and Istrefi (2023)) finds that board governors are generally more dovish than bank presidents. Thus, one interpretation of our results is that if the situation is such that even board governors voiced a tighter policy stance in the first round, then all members would be relatively more hawkish in the second round. Similarly, if the situation is such that even bank presidents voiced a less tight policy stance in the first round, then all members would be relatively more dovish in the policy round. This suggests that when FOMC members have different and known biases (some hawkish, others dovish), members are rational in the way that they incorporate information from the economic round: While they may discount information from another member if it aligns with that member’s bias, they give greater weight to the views of members who surprise the committee by contradicting their usual bias.

The previous results are consistent with those concerning a loose policy stance in columns four through six.²¹ That is, if board governors voiced a tighter policy stance in the first round, then members are less likely to voice a looser policy stance in the second round. Note, however, that only the coefficient for the chair is significant at the 5% level. If bank presidents voiced a looser policy stance in the first round, then all members are more likely to voice a looser stance in the second round, with the coefficients of board governors and bank presidents significant at the 5% level. In addition, a less tight policy stance by bank presidents and a looser stance by the chair in the first round predict a looser policy stance by both board governors and bank presidents in the second round, with the coefficients of the former significant at the 10% level and those of the latter significant at the 5%.

Table 5 shows that tone in the economic round has no predictive power over either the voiced tighter or looser policy stances in the policy round. In contrast, economic data has strong predictive power with higher inflation, lower unemployment, and lower housing starts forecasting a tighter policy stance by all members, except for the chair in the case of housing starts, and (surprisingly) a looser policy stance by bank presidents too. In what follows, we explore further the finding that higher inflation predicts both tighter and looser policy stances by bank presidents in the second

²¹To interpret this results, recall that we aggregate all board governors and all bank presidents into a single category each, and thus the looser and tighter policy stances may have been voiced by different individuals in the transcripts.

round. Figure 6 plots the relation between the voiced stances and inflation with full red dots denoting the high inflation period in the early 1980s. The positive slope in both figures accounts for the positive coefficients in Table 4 under both stances, and shows that they are driven by events in the high inflation period. Given that our measure of stance aggregates members of the same type, this result simply shows that bank presidents (and board governors) are not monolithic in their views and so during this high inflation period, bank presidents voiced both tighter and looser policy stances. For example, during this period bank presidents dissented from the final decision both in favor of a tighter policy (e.g., Guffey in May and September 1980) and a looser policy (e.g., Roos in May 1980 and February 1981).

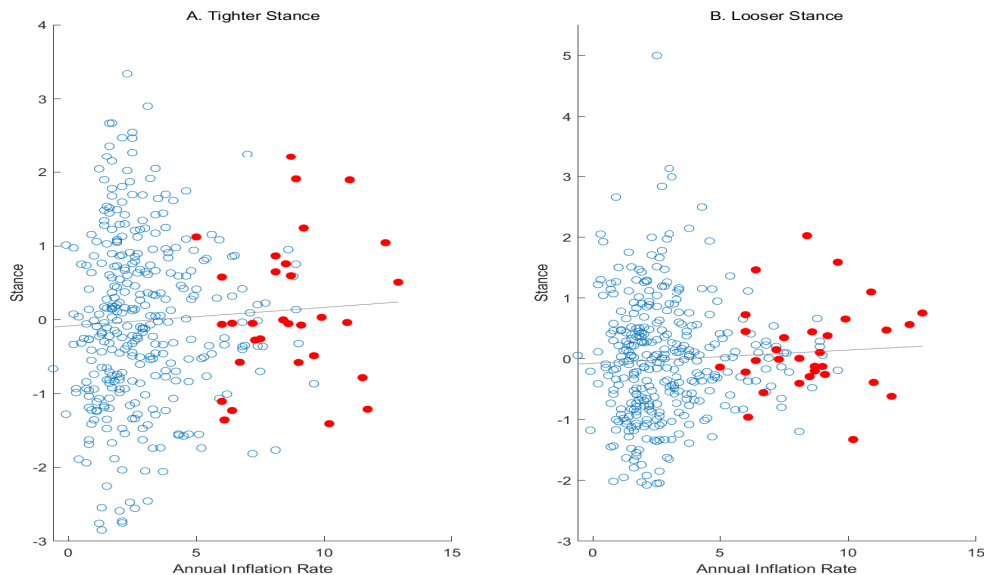


Figure 6: Relation Between Inflation and Bank Presidents' Voiced Policy Stance in Policy Round

Finally, regarding tone, a more positive tone in the first round of the meeting predicts a more positive tone, but less collaboration, in the second round; and more collaboration in the first round predicts more collaboration in the second round. The policy stances voiced by bank presidents during the economic round significantly predicts collaboration in subsequent policy round. Both tighter and looser policy stances have positive predictive power, but the coefficient of the former is twice the magnitude of the coefficient of the latter. Economic conditions also have strong predictive power on collaboration: lower output growth, lower inflation, higher unemployment, and an increase

in housing starts all predict more collaboration in the policy round. This is roughly in line with the results for the first round (though there is a change in the sign of the inflation coefficient). Interestingly, positive emotion in the policy round correlates positively with housing starts, but no other economic variable shows statistical significance in relation to positive emotion. Negative emotion in the first round has no predictive power over any of the variables examined, and no variable significantly predicts a negative tone in the second round.

In summary, these results show that deliberation in the economic round of the meeting affects deliberation in the second round, with voiced stances in the latter round forecastable on the basis of stances in the former round and by current economic conditions. Most notably, when board governors voice a tighter policy stance in the economic round, this predicts a shift towards tighter stances by all committee members in the policy round. Conversely, when bank presidents voice a looser policy stance in the economic round, this predicts a shift towards looser stances by all members in the policy round. One implication of these results is that the chair’s policy stance can be affected by the views expressed by other committee members as part of the deliberation.²² While tone and collaboration are persistent across rounds, they have no predictive power over policy stances in the policy round.

5.3 Across Meetings

Table 6 reports estimates of the deliberation parameters to answer the question of whether the voiced policy stance by the chair, board governors, and bank presidents, and the tone of the meeting can be predicted by the stances and tone in the previous meeting, controlling for current and past economic conditions.

An important result is that the dynamics of deliberation across meetings are different from those between rounds of the same meeting. For instance, Table 6 shows persistence in the voiced stances for tighter and looser policies within class memberships, with most coefficients significant at the 5% level and only the voiced stance for looser policy by board governors not statistically significant. This means that, even after controlling for the information contained in the Greenbooks, voiced stances carry over across meetings, possibly reflecting individual policy preferences, which

²²See, however, President Poole (quoted in Schonhardt-Bailey and Bailey (2013)), who stated “I’m pretty sure that Greenspan, in almost all cases, knew exactly the result he wanted, and his views were not affected by the meeting discussion.”

by definition are persistent—or, on a less positive note, they could also reflect rigidity (Laarits et al. (2025)) or unwillingness to change positions for fear of appearing incompetent (Prendergast and Stole (1996)). However, since significant autoregressive coefficients range from 0.13 to 0.32 and the other variables are significant, current stances are not mechanically determined by individual past stances but evolve as a result of deliberation.

For instance, a voiced stance for a looser policy by bank presidents in the previous meeting predicts more muted stances for a tighter policy, and more vocal stances for a looser policy, by both board governors and bank presidents in the current meeting. Similarly, a voiced stance for a tighter policy by bank presidents in the previous meeting predicts more muted stances for a looser policy by both board governors and bank presidents. These results suggest that bank presidents are exceptionally influential on the policy stance of members, other than the chair, across meetings. Two possible mechanisms explain this result. First, bank presidents frequently give speeches and interviews in the intra-meeting period, where they state their policy views, sometimes in stronger language than that used at the meetings. Ehrmann et al. (2021) examine the speech behavior of bank presidents, and find that it is related to voting status and regional economic conditions, suggesting that speeches may influence policy as other bank presidents and board governors hear the speech or read its transcript. Notably, because bank presidents interact less than board governors outside meetings, a president’s speech may convey new information to other FOMC members. Second, some of the regional information presented by bank presidents early in the meeting may constitute new information from the perspective of the other members despite the fact that regional information is circulated before the meeting in the form of a Beige book.²³ Along these lines, Governor Rivlin (quoted in p. 390 of Schonhardt-Bailey and Bailey (2013)) states “The FOMC meeting, from my point of view, was an opportunity to hear what the Bank presidents had to say about what was going on in their area, and that was the most interesting part of the meeting. I already knew what my fellow board governors thought because we had been interacting a lot.”

Interestingly, Table 6 shows that a looser policy stance by board governors in the previous meeting predicts a less tight policy stance by the chair in the current meeting. That is, while the previous stance of bank presidents has no effect on the chair, the previous stance of board governors

²³The Beige book summarizes anecdotal information on current economic conditions in each district, collected through reports from bank and branch directors and interviews with business contacts, economists, market experts, and other sources.

has some effect on the chair. One possibility is that this effect arises from the interaction between board governors and the chair in the period between meetings. See, for example, the quote from Governor Rivlin above, and Meyer (2006), who writes (p. 50): “To ensure [Greenspan] has the votes to support his policy recommendation, the Chairman visits with the members of the Board in advance of FOMC meetings.”

There is some effect of the tone in the previous meeting on the current stance by all members. In general, more collaboration mutes the stances for both a tighter and looser policy by all members, with the negative coefficient statistically significant in three of the six cases. One interpretation of this result is that past collaboration reduces polarization among committee members, inducing them to voice less strong policy views in the current meeting. Past negative emotion (associated, for example, with fear) in the previous meeting predicts a less tight policy stance by bank presidents and looser policy stances by all members in the current meeting. Regarding economic conditions, an increase in current output growth, a decrease in current unemployment, and an increase in lagged inflation predict a statistically significant tighter policy stance by some members. Likewise, a decrease in current output growth and increase in lagged unemployment predict a statistically significant looser policy stance by some members.

There is little effect of past policy stances on the current tone of the meeting. However, there is substantial persistence in tone across meetings with the coefficients of past negative emotion, positive emotion, and collaboration equal to 0.25, 0.16, and 0.59, respectively. In addition, positive emotion in the previous meeting reduces negative emotion in the current meeting, although the coefficient is only significant at the 10% level. Interestingly, positive emotion and collaboration exhibit an inverse relationship: A higher positive tone in the previous meeting predicts less collaboration in the current meeting, and conversely, higher previous collaboration predicts a lower current positive tone. Among economic conditions, only output growth shows a significant effect.

5.4 Granger Causality

As a summary of the interaction between tone and policy stance in FOMC meetings, we perform Granger causality tests on the vector autoregressions implied by our deliberation model, both within and across meetings, and report results in Table 7. The test statistic of this block exogeneity test is distributed χ^2 with degrees of freedom equal to the number of restrictions tested.

Regarding deliberation within the meeting, the hypothesis that the policy stance of members in the economic round does not help forecast the stance and tone in the policy round can be rejected at standard significance levels. The hypothesis that the tone in the economic round does not help forecast the tone in the policy round can be rejected as well. In contrast, the hypothesis that the tone in the economic round does not help forecast the policy stance in the policy round cannot be rejected (p -value = 0.815). The latter result is consistent with the results from Table 5, which show that none of the individual components of tone (negative emotion, positive emotion, and collaboration) in the economic round is statistically significant in explaining policy stances in the policy round. We conclude that the policy stance within FOMC meetings is block-exogenous with respect to the tone of the meeting.

Regarding deliberation across meetings, the hypothesis that the policy stance of members (resp. tone) in a meeting does not help forecast the stance (resp. tone) in the next meeting can be rejected at standard significance levels. Conversely, the hypothesis that the policy stance of members (resp. tone) in a meeting does not help forecast the tone (resp. stance) in the next meeting cannot be rejected at standard levels. This is so even though some individual coefficients in Table 5 are statistically different from zero. Therefore, we conclude that policy stance across FOMC meetings is block-exogenous with respect to the tone of the meeting, and tone across FOMC meetings is block-exogenous with respect to the policy stance.

Appendix A examines the robustness of our estimates to the measures of voiced policy stance, the economic variables from the Greenbooks, and a break in the data in October 1993, when the existence of transcripts became known. Except for a few instances where the statistical significance of a coefficients goes above or below standard levels, usually preserving the original sign, the estimates and main conclusions drawn from the benchmark results are generally robust to these three alternative specifications.

5.5 Implications of Transparency

We now examine in more detail the implications of transparency for deliberation. There are several reasons why we may expect deliberation to be affected by transparency. As Alan Greenspan argued in front of the U.S. Congress, as a result of transparency, “the candid airing of views, the forthright give and take, and the tentative posing of new ideas would likely disappear” (October 13, 1993),

while Janet Yellen expressed concerns that the exposure of differences of opinion among members could affect the degree of collegiality within the FOMC.²⁴ Transparency may also alter members’ incentives to voice their opinions for fear of appearing incompetent (Meade and Stasavage (2008)) or if they anticipate they will be held publicly accountable at some point in the future.²⁵

We study the effect of transparency by means of a sharp regression discontinuity design (RDD) where treatment is defined as FOMC members becoming aware in October 1993 that meeting transcripts were produced and kept by Federal Reserve.²⁶ Shortly thereafter, pressured by the U.S. Congress, the Federal Reserve decided to make past transcripts available and to release all future transcripts with a five-year delay. Note that the treatment was unanticipated and exogenous, and, so the assumption of ignorability for the RDD holds.

Under the maintained assumption that the underlying mean functions are continuous, the average treatment effect (ATE) is estimated by means of a local linear regressions in a window of four years (32 meetings) around October 1993, controlling for the Greenbook observations of current output growth, inflation, unemployment, and housing starts. Column 2 of Table 8 reports the coefficient of a dummy variable that takes value one for meetings after and including that of November 1993, and zero for meeting before.²⁷ In the language of RDD, the former are “treated” meetings while the latter are “untreated.” The dependent variable in each regression is noted in Column 1.

Estimates in Panels A and B indicate that after October 1993, members are less likely to voice a looser policy stance in the economic round—statistically significant in the case of the chair and bank presidents—and more likely to voice a tighter policy stance in the policy round—statistically significant in the case of governors and bank presidents. A possible explanation for this increase in hawkishness could be a desire to emulate more hawkish members if such members have a reputation for competence, or else it could reflect an effort to please political principals—for example, all board governors at the time had been appointed by Republican administrations, which traditionally are

²⁴Speech at the Bank of Japan on May 31, 2005.

²⁵This is more than just a theoretical possibility. For instance, during a Senate hearing in 2014, Senator Brown asked Chair Yellen why, “on the eve of the Lehman bankruptcy, FOMC members mentioned, according to the [transcripts], inflation 129 times and recession 5 times.”

²⁶The timing of events is as follows. Several FOMC members learned about the existence of transcripts from Greenspan on October 15 during a conference call. Greenspan testified before Congress on October 19 and, although he did not mention the existence of transcripts during this hearing, he admitted to it in a letter to the House Committee on October 26. See *The Federal Reserve’s 17-Year Secret* prepared by the staff of the Committee on Banking, Finance, and Urban Affairs of the U.S. House of Representatives published on January 1994 and available from the website of the Federal Reserve Bank of St. Louis (<https://www.stlouisfed.org/>)

²⁷The other coefficients are not reported to save space but are available from the authors upon request.

deemed to be more concerned about price stability. Interestingly, transparency after October 1993 induces a statistically significant decline in voiced negative emotions (anger, disgust, fear, and sadness) in both rounds of the meeting, and an increase in positive emotions in the policy round. Hence, the knowledge that their interventions will be made eventually available to the public, affects the tone of the meeting and voice policy stances by members. These results complement those in Hansen et al. (2018), who report that transparency also affects other dimensions of the meeting leading to longer, more quantitative, and more technical interventions by members, with substantial differences across experience. Moreover, our findings connect to those of Iaryczower et al. (2025), who find that transparency makes governors, but not bank presidents, less dovish. Their results are based on numerical policy recommendations from the policy round, whereas we rely on text-based measures of policy stance in both rounds.

Estimates in Panel C indicate that after October 1993, there is a statistically significant decrease in the probability that board governors dissent—i.e., that they officially register in the transcripts their disagreement with the policy selected by the committee—and, hence, their dissents have become relatively uncommon.²⁸ In contrast, transparency has no effect on the dissent behavior of bank presidents. Our finding that dissent decreases with transparency is consistent with Meade and Stasavage (2008). However, in contrast to our paper, they primarily analyze voiced dissents inferred from transcripts and find little effect when examining official dissents.

6 Policy Outcomes

We now examine the effect of the deliberative process on two outcomes, namely the federal funds rate target selected by the committee and the decision to dissent on the part of FOMC members.

6.1 Interest Rate Decisions

Table 9 reports OLS estimates of the deliberation policy rule (12), which incorporates measures of voiced policy stance and tone for the whole meeting, in addition to the Greenbook variables. For the purpose of comparison, the table also reports OLS estimates of a standard Taylor-type rule. As discussed in Section 4.1, based on Coibion and Gorodnichenko (2012), we use an autoregressive

²⁸Indeed, this is one of the reasons why the recent dissents by governors since the summer of 2025 have attracted media attention. For instance, see *New York Times* August 2, 2025.

process of order higher than 1. In particular, we use an autoregressive process of order three to capture policy smoothing by the FOMC. A Durbin-Watson test of the hypothesis that the residuals of the regression are uncorrelated delivers a p -value of 0.739 and, thus, the hypothesis cannot be rejected. A Breusch–Pagan test of the hypothesis that the residuals are homoskedastic delivers a p -value less than 0.001 and, thus, the hypothesis can be rejected. For this reason, the standard errors reported in parenthesis in Table 9 are Huber–White heteroskedasticity-consistent standard errors.

The results show a statistically significant association between the bank presidents’ voiced stance for tighter policy and an interest rate target higher by 5.7 basis points (bps), and between the governors’ voiced stance for looser policy and an interest rate target lower by 5.3 bps. As a comparison, an increase in inflation by 1 percentage point induces a statistically significant increase in the target by 5.6 bps. These results are in line with previous literature based on formal dissents (e.g., Belden (1989)), expressed policy preferences (Eijffinger et al. (2015)), and news coverage (Istrefi (2018) and Bordo and Istrefi (2023)) that finds that governors are generally more dovish than bank presidents.

Neither negative nor positive emotions have predicted power over interest rate decisions, but the coefficient of collaboration is positive and statistically different from zero. As explained in Section 5.1, economic conditions affect the members’ willingness to collaborate and Figure 5 shows a positive association between collaboration and the inflation rate. Hence, we conclude that this association is driven by the high level of collaboration during Volcker’s chairmanship when large interest rate increases were approved by the FOMC.

Overall, our results show that textual measures of voiced policy stance and tone help forecast FOMC policy decisions, above and beyond real-time Greenbook variables.²⁹

6.2 Dissent

In this section we study the relation between deliberation and dissent. We consider two measures of dissent. The first measure is whether there is any dissent by a board governor or a bank president

²⁹In related work, Bennani and Romelli (2024) use a bag-of-words approach to compute the tone of FOMC meetings and find that past tone has predictive power over changes in the federal funds rate target. See also Sharpe et al. (2023), Aruoba and Drechsel (2025)), and other literature discussed in the Introduction that incorporates sentiment measures into reaction functions.

(separately) in a given meeting. This is a binary variable taking value 1 if there is a dissent and 0 otherwise. The second measure is the dissent rate by board governors and bank presidents (separately), measured by the number of dissents divided by the total number of FOMC members present at the meeting. This is a censored variable taking only non-negative values. Respective results from Probit and Tobit regressions relating these dissent measures to the voiced policy stance and tone in the policy round of the meeting and economic variables are reported in Table 10.

The probability of a dissent and dissent rate by bank presidents increases when they have voiced a tighter policy stance during the meeting. This result is intuitive and in line with the dissent data collected by Thornton and Wheelock (2014), which for our sample imply that 84% of all dissents by bank presidents were for a tighter policy than that selected by the committee. The probability of dissent and dissent rate by board governors increase with the rate of inflation and housing starts. Thornton and Wheelock (2014) also report a weak, but statistically significant, relation between the dissent rate and inflation, and our results in Tables 8 and 10 suggest that their finding is primarily driven by the behavior of board governors prior to 1993.³⁰

Finally, an important result in Table 10 is that the probability of dissent and dissent rate by both governors and bank presidents increases with the level of negative emotions (anger, disgust, fear, and sadness) in the meeting and decreases with the level of positive emotions (happiness). Thus, while results in Tables 7 and 9 show that tone does not Granger-cause voiced policy stances in the meeting and (except for collaboration) does not help forecast interest rate decision, tone—in particular, negative and positive emotions—plays a role in the willingness of FOMC members to formally register their dissent at the end of the meeting. This result is important because previous research by Riboni and Ruge-Murcia (2014) shows that current dissents help forecast future policy decisions by the FOMC.³¹ That is, a current dissent in favor of, say, an interest rate increase is a statistically significant predictor of an interest rate increase in the FOMC’s next meeting.

³⁰In recent work, Tsang and Yang (2025) estimate a measure of hidden dissent (or disagreement) using a deep language model and find that it is strongly related to inflation and much less so to unemployment.

³¹See also Gerlach-Kristen (2009), who reports similar evidence for the Monetary Policy Committee of the Bank of England.

7 Conclusions

A key challenge in studying deliberation at FOMC meetings is to quantify the exchanges among committee members and their influence on decision-making. To overcome this challenge, this paper uses artificial intelligence and natural language processing to extract measures of policy stance, emotions, and collaboration from the transcripts of FOMC meetings. These measures are then used as input in an econometric model of deliberation where members interact across rounds of the same meeting and over time across meetings. The model permits the interaction of policy stance, emotions, and collaboration between members, and its estimation reveals that FOMC members learn from one another during within-meeting deliberation and give greater weight to opinions that are surprising relative to the source member’s usual stance. Members also influence one another across meetings, with bank presidents being exceptionally influential. Block-exogeneity tests indicate (perhaps, reassuringly) that emotional tone has a limited effect on policy stances and interest rate decisions. However, emotional tone has predictive power over dissent behavior, with a more negative tone increasing the probability that a member dissents from the decision made by the committee.

Overall, our findings show that FOMC decisions are not simply the average of the members’ preferences. Instead, they reflect an exchange of information that can contribute to improve policy making, in line with the experimental findings of Blinder and Morgan (2008) and Lombardelli et al. (2005). Some of our results regarding how information is incorporated are consistent with rational behavior by FOMC members. However, our findings also suggest that FOMC decision-making is a collaborative process, with interpersonal dynamics and behavioral factors playing some role, and consistent with Janet Yellen’s observation that “FOMC members behave far less individualistically and strategically than assumed in some [economic] models.”³²

³²Presentation at the Bank of Japan on May 31, 2005.

Table 1: Proportion of Concurrent Voiced Tighter and Looser Preferences

	Rule Based	Language Model
Whole meeting	94.13	100.00
Economic round	88.55	99.44
Policy round	66.48	99.16

Note: This table reports the proportion of FOMC meetings, or rounds of FOMC meetings, where both voiced preferences for tighter and looser preferences are expressed.

Table 2: Bartlett Tests

	Tighter Stance	Looser Stance	Tone		
			Negative	Positive	Collab.
A. Chair					
Whole meeting	0.000	0.000	0.105	—	0.000
Economic round	0.000	0.000	0.000	—	0.000
Policy round	0.000	0.000	0.000	—	0.000
B. board governors					
Whole meeting	0.000	0.000	0.000	—	0.000
Economic round	0.000	0.000	0.000	—	0.000
Policy round	0.000	0.013	0.009	—	0.000
C. Bank Presidents					
Whole meeting	0.000	0.000	0.000	—	0.000
Economic round	0.000	0.000	0.000	—	0.000
Policy round	0.000	0.000	0.000	—	0.000

Note: This table reports the p -values of Bartlett tests of the hypothesis that the correlation matrix of the series is the identity matrix.

Table 3: Correlation of Voiced Policy Stances

	Tight Chair	Tight Board	Tight Bank	Loose Chair	Loose Board	Loose Bank
Tight chair	1.000	0.573	0.540	0.258	-0.025	-0.059
Tight board	0.573	1.000	0.781	-0.035	0.050	-0.077
Tight bank	0.540	0.781	1.000	-0.062	-0.119	-0.138
Loose chair	0.258	-0.035	-0.062	1.000	0.253	0.369
Loose board	-0.025	0.050	-0.119	0.253	1.000	0.620
Loose bank	-0.059	-0.077	-0.138	0.369	0.620	1.000

Note: This table reports the correlation of voiced policy stances for the three member types for the whole meeting.

Table 4: Deliberation in Economic Round

	Regressor				
Response	Output			Housing	
Variable	Growth	Inflation	Unemployment	Starts	R^2
A. Tighter Stance					
Chair	−0.006 (−0.206)	0.028 (1.261)	−0.002 (−0.063)	−0.175 (−0.909)	0.009
Board	0.105 (3.618)	0.023 (0.862)	−0.207 (−5.457)	0.191 (1.033)	0.200
Bank	0.160 (5.250)	0.022 (0.818)	−0.229 (−6.909)	0.193 (0.983)	0.301
B. Looser Stance					
Chair	−0.040 (−1.513)	−0.018 (−0.973)	0.164 (3.924)	−0.276 (−1.421)	0.109
Board	−0.122 (−3.941)	−0.040 (−1.883)	0.156 (3.793)	−0.098 (−0.525)	0.147
Bank	−0.171 (−4.719)	−0.047 (−1.529)	0.139 (3.408)	0.187 (0.792)	0.137
C. Tone					
Negative	−0.133 (−3.987)	−0.061 (−2.121)	0.091 (2.089)	0.298 (1.403)	0.065
Positive	0.011 (0.546)	−0.050 (−2.718)	−0.038 (−1.000)	0.503 (3.016)	0.097
Collab.	−0.089 (−2.513)	0.080 (2.100)	0.057 (1.156)	−0.254 (−1.122)	0.112

Note: The figures in parenthesis are t statistics.

Table 5: Deliberation in Policy Round

Regressor	Response Variable								
	Tight	Tight	Tight	Loose	Loose	Loose	Tone		
	Chair	Board	Bank	Chair	Board	Bank	Negative	Positive	Collab.
Intercept	0.863 (2.266)	1.550 (3.993)	2.047 (5.016)	0.163 (0.459)	0.631 (1.536)	0.880 (2.802)	0.100 (0.270)	-0.429 (-1.396)	-1.608 (-3.460)
Lag tight chair	-0.101 (-1.839)	0.054 (0.889)	0.022 (0.314)	-0.101 (-1.484)	-0.037 (-0.721)	0.035 (0.746)	0.078 (1.364)	-0.029 (-0.567)	0.119 (1.702)
Lag tight board	0.313 (3.817)	0.243 (2.857)	0.142 (1.778)	-0.173 (-2.450)	-0.106 (-1.506)	-0.082 (-1.386)	-0.095 (-1.105)	0.045 (0.640)	-0.190 (-1.877)
Lag tight bank	-0.090 (-1.012)	-0.057 (-0.548)	0.056 (0.539)	0.085 (1.043)	-0.131 (-1.668)	-0.120 (-1.709)	-0.051 (-0.523)	-0.153 (-1.710)	0.327 (3.063)
Lag loose chair	-0.062 (-1.026)	-0.062 (-1.156)	0.043 (0.727)	-0.017 (-0.359)	0.126 (2.076)	0.122 (2.267)	0.065 (1.452)	-0.070 (-1.508)	-0.078 (-1.584)
Lag loose board	0.027 (0.409)	-0.066 (-1.026)	-0.088 (-1.332)	-0.048 (-0.784)	-0.013 (-0.208)	0.083 (1.047)	-0.042 (-0.603)	-0.001 (-0.030)	-0.121 (-1.566)
Lag loose bank	-0.159 (-2.319)	-0.127 (-1.821)	-0.220 (-3.074)	0.117 (1.613)	0.117 (2.015)	0.186 (3.530)	0.086 (1.432)	0.033 (0.775)	0.185 (2.622)
Lag negative	-0.059 (-1.009)	-0.065 (-1.043)	-0.047 (-0.825)	0.039 (0.567)	0.085 (1.413)	0.023 (0.428)	0.011 (0.200)	-0.077 (-1.515)	-0.039 (-0.609)
Lag positive	-0.048 (-0.607)	-0.013 (-0.198)	0.056 (0.792)	0.052 (0.748)	-0.019 (-0.359)	0.037 (0.583)	-0.048 (-0.779)	0.223 (2.030)	-0.130 (-1.521)
Lag collaboration	0.005 (0.103)	0.083 (1.504)	0.052 (0.993)	-0.065 (-1.566)	0.015 (0.319)	0.035 (0.840)	-0.015 (-0.276)	-0.086 (-2.212)	0.346 (5.334)
Output growth	0.001 (0.043)	0.017 (0.563)	0.028 (0.956)	-0.033 (-1.282)	0.026 (1.084)	0.020 (0.865)	0.016 (0.545)	-0.011 (-0.462)	-0.089 (-2.900)
Inflation	0.050 (2.198)	0.072 (2.783)	0.080 (2.987)	0.026 (1.195)	0.028 (1.343)	0.044 (2.477)	0.008 (0.356)	-0.007 (-0.362)	-0.057 (-2.122)
Unemployment	-0.140 (-3.496)	-0.198 (-4.736)	-0.201 (-4.710)	-0.030 (-0.799)	-0.060 (-1.407)	-0.072 (-2.114)	0.008 (0.194)	-0.018 (-0.586)	0.211 (4.256)
Housing starts	-0.102 (-0.533)	-0.410 (-2.115)	-0.793 (-3.826)	0.016 (0.084)	-0.291 (-1.488)	-0.442 (-2.780)	-0.154 (-0.835)	0.427 (2.273)	0.480 (2.269)
R^2	0.194	0.212	0.243	0.062	0.148	0.240	0.059	0.125	0.211

Note: The figures in parenthesis are t statistics.

Table 6: Deliberation Across Meetings

Regressor	Response Variable								
	Tight	Tight	Tight	Loose	Loose	Loose	Tone		
	Chair	Board	Bank	Chair	Board	Bank	Negative	Positive	Collab.
Intercept	0.962 (2.433)	0.799 (2.141)	0.738 (2.191)	-0.223 (-0.575)	0.298 (0.834)	0.356 (1.064)	0.664 (1.701)	-0.270 (-0.767)	-0.736 (-1.988)
Lag tight chair	0.129 (1.886)	-0.019 (-0.310)	-0.036 (-0.727)	0.067 (0.949)	-0.005 (-0.096)	0.004 (0.066)	0.016 (0.248)	-0.036 (-0.729)	-0.062 (-1.418)
Lag tight board	0.058 (0.704)	0.300 (3.903)	0.081 (1.192)	-0.096 (-1.215)	-0.087 (-1.274)	-0.029 (-0.455)	0.089 (1.072)	0.013 (0.184)	0.092 (1.171)
Lag tight bank	0.027 (0.318)	0.037 (0.456)	0.185 (2.555)	0.016 (0.186)	-0.167 (-2.247)	-0.185 (-2.822)	-0.190 (-2.173)	-0.013 (-0.218)	0.126 (1.604)
Lag loose chair	0.063 (0.965)	0.076 (1.384)	0.031 (0.707)	0.137 (2.086)	0.007 (0.132)	0.022 (0.474)	0.019 (0.346)	0.005 (0.110)	-0.052 (-1.036)
Lag loose board	-0.203 (-2.915)	-0.033 (-0.449)	-0.065 (-1.135)	0.088 (1.437)	0.059 (0.780)	0.098 (1.239)	0.048 (0.523)	-0.037 (-0.696)	0.092 (1.587)
Lag loose bank	-0.076 (-0.970)	-0.174 (-2.648)	-0.192 (-3.814)	-0.058 (-0.762)	0.146 (1.833)	0.317 (5.296)	0.128 (1.505)	0.057 (1.092)	-0.008 (-0.146)
Lag negative	0.037 (0.571)	-0.031 (-0.558)	-0.134 (-2.746)	0.138 (1.980)	0.130 (2.382)	0.199 (4.010)	0.252 (4.127)	-0.015 (-0.386)	-0.017 (-0.344)
Lag positive	0.063 (0.886)	0.027 (0.350)	0.010 (0.164)	0.050 (0.805)	0.117 (1.342)	0.002 (0.027)	-0.122 (-1.840)	0.159 (2.516)	-0.193 (-3.201)
Lag collaboration	-0.067 (-1.399)	-0.053 (-1.183)	-0.076 (-1.835)	-0.083 (-1.908)	-0.096 (-2.146)	-0.015 (-0.410)	-0.013 (-0.272)	-0.125 (-3.212)	0.590 (10.471)
Output growth	0.052 (1.146)	0.060 (1.702)	0.075 (2.026)	-0.118 (-2.821)	-0.027 (-0.645)	-0.076 (-2.122)	-0.093 (-1.797)	0.049 (1.677)	0.035 (0.821)
Inflation	-0.054 (-0.894)	-0.041 (-0.748)	0.020 (0.413)	0.027 (0.514)	-0.006 (-0.094)	-0.003 (-0.050)	0.064 (0.935)	-0.031 (-0.861)	-0.014 (-0.263)
Unemployment	0.003 (0.013)	-0.378 (-2.306)	-0.257 (-1.505)	0.251 (1.449)	0.372 (1.642)	0.291 (1.625)	0.263 (1.015)	0.015 (0.106)	-0.026 (-0.116)
Housing starts	-1.067 (-1.589)	-0.065 (-0.116)	-0.408 (-0.784)	0.265 (0.433)	-0.855 (-1.428)	0.260 (0.523)	-0.518 (-0.871)	0.303 (0.644)	-0.632 (-1.005)
Lag output growth	-0.038 (-0.980)	-0.045 (-1.356)	-0.045 (-1.302)	0.093 (2.732)	0.001 (0.014)	0.070 (2.232)	0.098 (2.137)	-0.044 (-1.737)	-0.099 (-2.411)
Lag inflation	0.109 (1.799)	0.090 (1.597)	0.015 (0.291)	-0.027 (-0.488)	0.031 (0.433)	0.020 (0.355)	-0.055 (-0.751)	0.004 (0.095)	0.020 (0.357)
Lag unemployment	-0.122 (-0.577)	0.216 (1.305)	0.089 (0.519)	-0.194 (-1.175)	-0.365 (-1.661)	-0.348 (-1.948)	-0.302 (-1.175)	-0.047 (-0.333)	0.140 (0.635)
Lag Housing starts	0.766 (1.127)	0.093 (0.157)	0.509 (0.977)	-0.316 (-0.489)	0.599 (1.013)	-0.283 (-0.548)	0.196 (0.326)	0.094 (0.198)	0.734 (1.146)
R^2	0.196	0.404	0.506	0.155	0.325	0.475	0.303	0.199	0.479

Table 7: Granger Causality Tests

Null Hypothesis	χ^2 Statistic	Degrees of Freedom	p -value
<i>A. Within Meeting</i>			
Policy stance in the economic round does not help forecast:			
Policy stance in the policy round	149.37	36	<0.001
Tone in the policy round	41.18	18	0.001
Tone in the economic round does not help forecast:			
Policy stance in the policy round	12.60	18	0.815
Tone in the policy round	53.35	9	<0.001
<i>B. Across Meetings</i>			
Policy stance in the previous meeting does not help forecast:			
Policy stance in the current meeting	58.292	18	<0.001
Tone in the current meeting	11.451	18	0.874
Tone in the previous meeting does not help forecast:			
Policy stance in the current meeting	17.22	18	0.508
Tone in the current meeting	160.03	9	<0.001

Table 8: Average Treatment Effects

Variable	ATE	<i>t</i> Statistic
<i>A. Economic Round</i>		
Tighter stance		
Chair	-0.212	(-0.518)
Board	-0.151	(-0.242)
Bank	0.141	(0.266)
Looser stance		
Chair	-0.780	(-1.933)
Board	-0.510	(-1.023)
Bank	-1.006	(-2.286)
Tone		
Negative	-1.363	(-3.889)
Positive	0.226	(0.693)
Collaboration	0.029	(0.102)
<i>B. Policy Round</i>		
Tighter stance		
Chair	0.222	(0.465)
Board	1.220	(2.359)
Bank	0.928	(2.112)
Looser stance		
Chair	0.480	(1.153)
Board	0.079	(0.173)
Bank	0.521	(0.986)
Tone		
Negative	-1.168	(-2.052)
Positive	1.312	(3.272)
Collaboration	-0.832	(-2.115)
<i>C. Dissent</i>		
Whether to dissent		
Board	-2.413	(-2.564)
Bank	-0.680	(-0.996)
Dissent rate		
Board	-0.316	(-2.092)
Bank	-0.073	(-0.774)

Note: This table reports estimates of the ATE of transparency on the variables in the first column. Estimates were obtained using local linear regressions (Panels A and B) and local Probit and Tobit regressions (Panel C).

Table 9: Deliberation and Interest Rate Decisions

Regressor	Deliberation		Taylor Rule	
	Estimate	<i>t</i> Statistic	Estimate	<i>t</i> Statistic
Intercept	-0.118	(-0.530)	-0.190	(-0.890)
Interest rate lags				
I_{t-1}	1.140	(14.063)	1.230	(15.644)
I_{t-2}	-0.051	(-0.369)	-0.068	(-0.474)
I_{t-3}	-0.118	(-1.377)	-0.177	(-1.703)
Tighter stance				
Chair	0.310	(1.252)		
Board	-0.174	(-0.591)		
Bank	0.572	(1.875)		
Looser stance				
Chair	-0.129	(-0.725)		
Board	-0.527	(-2.179)		
Bank	-0.258	(-0.955)		
Tone				
Negative	-0.387	(-1.890)		
Positive	0.071	(0.313)		
Collaboration	0.533	(2.878)		
Greenbook data				
Output growth	0.051	(0.174)	0.303	(1.066)
Inflation	0.562	(3.365)	0.343	(2.034)
Unemployment	0.098	(0.454)	-0.151	(-0.725)
Housing starts	0.103	(0.106)	1.047	(1.112)

Note: The coefficients (but not the *t* statistics) reported for stance, emotion, and Greenbook data have been multiplied by 10 for readability.

Table 10: Deliberation and Dissent

Regressor	Whether to Dissent		Dissent Rate	
	Board	Bank	Board	Bank
Intercept	−3.875 (−5.104)	−1.214 (−2.105)	−0.545 (−4.730)	−0.127 (−1.726)
Tight board	0.018 (0.152)	−0.011 (−0.124)	0.001 (0.051)	0.001 (0.062)
Tight bank	0.207 (1.788)	0.263 (2.900)	0.020 (1.299)	0.033 (2.882)
Loose board	0.125 (1.217)	0.032 (0.407)	0.016 (1.158)	0.002 (0.247)
Loose bank	0.094 (0.799)	0.096 (1.114)	0.010 (0.653)	0.012 (1.054)
Negative	0.223 (2.451)	0.171 (2.182)	0.026 (2.177)	0.020 (2.099)
Positive	−0.580 (−3.270)	−0.575 (−4.396)	−0.078 (−3.280)	−0.077 (−4.514)
Collaboration	0.209 (2.522)	−0.043 (−0.594)	0.024 (2.208)	−0.002 (−0.219)
Output growth	0.056 (1.160)	−0.031 (−0.776)	0.007 (1.089)	−0.002 (−0.428)
Inflation	0.236 (5.575)	−0.025 (−0.721)	0.029 (5.530)	0.000 (0.075)
Unemployment	0.085 (1.159)	0.106 (1.805)	0.013 (1.388)	0.010 (1.427)
Housing starts	0.845 (2.315)	0.106 (0.386)	0.132 (2.737)	−0.001 (−0.026)

Note: The figures in parenthesis are t statistics.

A Appendix A: Robustness

This appendix examines the robustness of our estimates to 1) the measures of voiced policy stance, 2) the economic variables from the Greenbooks, and 3) a break in the data in October 1993, when the existence of transcripts became known to FOMC members. First, as an alternative measure of the voiced policy stance we consider

$$\textit{tighter stance rule } (i, t) = \log(1 + \#\textit{tighter pairs in } (i, t)),$$

$$\textit{looser stance rule } (i, t) = \log(1 + \#\textit{looser pairs in } (i, t)),$$

for the rule-based measure and

$$\textit{tighter stance language } (i, t) = \log(1 + \#\textit{tighter labels in } (i, t)),$$

$$\textit{looser stance language } (i, t) = \log(1 + \#\textit{looser labels in } (i, t)).$$

for the stance constructed using the language model. As before, our measure of tighter policy stance is the first principal component of both tighter measures, and likewise for the looser policy stance. Estimates of the deliberation model using these measures of policy stance are reported in Tables A.1 through A.3.

Second, instead of current values of output growth, inflation, unemployment, and housing starts, we employ the corresponding forecasts available from the Greenbooks. Note, however, that these forecasts are conditional on a policy path and, thus, they differ from market forecasts. Although the Greenbooks report forecasts at several horizons, the full set of forecasts for our variables and meetings is only available for one-, two-, and three-quarter ahead. Estimates of the deliberation model using forecasts at the three-quarter horizon are reported in Tables A.4 through A.6.

Finally, instead of a smooth quadratic trend to account for changes in English usage (Bauer (1994)), we allow for a break in a linear trend starting with the meeting of November 1993, which followed Alan Greenspan’s disclosure of the existence of FOMC meeting transcripts kept by the U.S. Federal Reserve. Estimates of the deliberation model allowing for a trend break in October 1993 are reported in Tables A.7 through A.9.³³

³³Section 5.6 in the text examines in detail the implications of this increase in transparency for the variables used in our analysis.

Comparing the estimates from these tables with the benchmark estimates in Tables 4 through 6, we can see that except for a few instances where the statistical significance of a coefficient goes above or below standard levels, usually preserving the original sign, the estimates and main conclusions drawn from the benchmark results are generally robust to these three alternative specifications. As an example consider Panel B in Tables 4, A.1, A.4, and A.7. In all cases, a rise in output growth or inflation (resp. unemployment) decreases (resp. increases) the looser policy stance of governors and bank presidents, and has no effect on the stance of the chair. However, the statistical significance of some of these coefficients depends on the specification. Likewise in Tables 5, A.2, A.5, and A.8, a voiced stance for a tighter policy by governors and looser policy stance by bank presidents in the economic round predicts the same stance by all members in the policy round, with a few instances where the statistical significance of the coefficient depends on the precise specifications.

Table A.1: Deliberation in Economic Round - Alternative Stance Measures

Response Variable	Regressor				R^2
	Output Growth	Inflation	Unemployment	Housing Starts	
A. Tighter Stance					
Chair	0.010 (0.436)	0.023 (1.388)	0.010 (0.313)	-0.304 (-1.660)	0.019
Board	0.089 (3.545)	0.035 (1.538)	-0.249 (-6.537)	0.216 (1.163)	0.226
Bank	0.138 (5.101)	0.026 (1.224)	-0.231 (-6.647)	0.285 (1.520)	0.302
B. Looser Stance					
Chair	0.002 (0.097)	-0.011 (-0.700)	0.182 (4.490)	-0.521 (-2.845)	0.172
Board	-0.107 (-3.724)	-0.025 (-1.165)	0.051 (1.248)	0.040 (0.197)	0.057
Bank	-0.123 (-3.935)	-0.036 (-1.389)	0.059 (1.557)	0.308 (1.489)	0.058
C. Tone					
Negative	-0.133 (-3.987)	-0.061 (-2.121)	0.091 (2.089)	0.298 (1.403)	0.065
Positive	0.011 (0.546)	-0.050 (-2.718)	-0.038 (-1.000)	0.503 (3.016)	0.097
Collab.	-0.089 (-2.513)	0.080 (2.100)	0.057 (1.156)	-0.254 (-1.122)	0.112

Note: The figures in parenthesis are t statistics.

Table A.2: Deliberation in Policy Round - Alternative Stance Measures

Regressor	Response Variable								
	Tight	Tight	Tight	Loose	Loose	Loose	Tone		
	Chair	Board	Bank	Chair	Board	Bank	Negative	Positive	Collab.
Intercept	1.030 (2.789)	1.403 (3.644)	1.906 (4.672)	0.643 (1.746)	0.952 (2.169)	1.215 (3.586)	0.096 (0.259)	-0.407 (-1.367)	-1.554 (-3.288)
Lag tight chair	-0.321 (-4.447)	0.184 (2.461)	0.108 (1.518)	-0.338 (-4.987)	-0.077 (-1.213)	0.071 (1.030)	0.017 (0.268)	0.005 (0.087)	0.169 (2.226)
Lag tight board	0.369 (4.419)	0.276 (3.308)	0.187 (2.511)	0.060 (0.737)	0.023 (0.290)	0.015 (0.183)	-0.054 (-0.618)	-0.047 (-0.744)	-0.070 (-0.810)
Lag tight bank	0.092 (0.985)	-0.047 (-0.475)	0.062 (0.658)	-0.064 (-0.706)	-0.210 (-2.300)	-0.280 (-3.442)	-0.133 (-1.253)	-0.091 (-1.105)	0.098 (1.016)
Lag loose chair	0.000 (0.000)	-0.173 (-2.720)	0.010 (0.151)	-0.056 (-0.927)	0.015 (0.186)	0.072 (0.990)	0.020 (0.351)	-0.097 (-1.869)	-0.105 (-1.751)
Lag loose board	-0.004 (-0.055)	0.030 (0.476)	-0.086 (-1.309)	0.082 (1.036)	0.045 (0.641)	0.027 (0.447)	0.014 (0.213)	-0.026 (-0.557)	-0.040 (-0.580)
Lag loose bank	-0.084 (-1.108)	-0.265 (-3.320)	-0.148 (-1.827)	0.146 (1.730)	0.029 (0.364)	0.300 (4.619)	0.037 (0.479)	0.068 (1.320)	0.010 (0.135)
Lag negative	-0.083 (-1.483)	-0.065 (-1.046)	-0.079 (-1.355)	-0.046 (-0.754)	0.076 (1.206)	-0.014 (-0.260)	0.011 (0.204)	-0.076 (-1.461)	-0.001 (-0.010)
Lag positive	-0.072 (-0.840)	-0.047 (-0.688)	0.056 (0.806)	0.030 (0.371)	-0.030 (-0.481)	0.041 (0.659)	-0.059 (-0.979)	0.220 (2.030)	-0.097 (-1.153)
Lag collaboration	0.015 (0.269)	0.135 (2.649)	0.069 (1.444)	-0.026 (-0.508)	0.128 (2.704)	0.046 (1.108)	-0.008 (-0.162)	-0.094 (-2.569)	0.320 (5.040)
Output growth	0.024 (0.974)	0.019 (0.631)	0.038 (1.409)	0.005 (0.188)	0.019 (0.692)	0.031 (1.251)	0.014 (0.507)	-0.012 (-0.486)	-0.080 (-2.622)
Inflation	0.037 (1.865)	0.062 (2.530)	0.066 (2.717)	0.033 (1.765)	0.037 (1.767)	0.055 (2.839)	0.010 (0.400)	-0.005 (-0.284)	-0.053 (-1.910)
Unemployment	-0.136 (-3.581)	-0.175 (-4.149)	-0.188 (-4.421)	-0.109 (-2.715)	-0.104 (-2.418)	-0.126 (-3.515)	0.004 (0.086)	-0.017 (-0.555)	0.192 (3.809)
Housing starts	-0.248 (-1.397)	-0.389 (-2.087)	-0.731 (-3.596)	-0.052 (-0.309)	-0.329 (-1.566)	-0.483 (-2.794)	-0.132 (-0.699)	0.404 (2.249)	0.502 (2.241)
R^2	0.318	0.240	0.204	0.177	0.100	0.232	0.055	0.124	0.180

Note: The figures in parenthesis are t statistics.

Table A.3: Deliberation Across Meetings - Alternative Stance Measures

Regressor	Response Variable								
	Tight	Tight	Tight	Loose	Loose	Loose	Tone		
	Chair	Board	Bank	Chair	Board	Bank	Negative	Positive	Collab.
Intercept	0.911 (2.385)	0.611 (1.511)	0.332 (0.947)	0.170 (0.416)	0.550 (1.364)	0.288 (0.830)	0.665 (1.722)	-0.286 (-0.793)	-0.635 (-1.681)
Lag tight chair	0.208 (2.762)	0.002 (0.023)	-0.023 (-0.442)	0.048 (0.733)	-0.091 (-1.231)	0.003 (0.053)	-0.008 (-0.113)	-0.000 (-0.002)	-0.004 (-0.089)
Lag tight board	0.099 (1.067)	0.353 (4.314)	0.054 (0.798)	-0.094 (-1.207)	0.116 (1.424)	-0.064 (-0.955)	0.119 (1.298)	-0.038 (-0.532)	0.012 (0.173)
Lag tight bank	0.053 (0.578)	0.036 (0.407)	0.363 (4.960)	0.022 (0.256)	-0.229 (-2.537)	-0.181 (-2.478)	-0.289 (-3.205)	0.020 (0.302)	0.092 (1.192)
Lag loose chair	0.023 (0.336)	0.011 (0.179)	0.027 (0.530)	0.091 (1.297)	0.140 (1.971)	0.080 (1.366)	0.029 (0.490)	-0.044 (-0.881)	-0.012 (-0.237)
Lag loose board	-0.148 (-2.059)	0.017 (0.260)	-0.090 (-1.754)	-0.034 (-0.508)	0.141 (1.979)	0.037 (0.560)	0.035 (0.499)	-0.022 (-0.421)	0.021 (0.442)
Lag loose bank	-0.156 (-1.968)	-0.208 (-2.995)	-0.162 (-2.998)	-0.022 (-0.265)	0.122 (1.298)	0.345 (4.959)	0.123 (1.753)	0.057 (1.126)	0.005 (0.093)
Lag negative	0.086 (1.538)	-0.046 (-0.819)	-0.104 (-2.258)	0.183 (3.217)	0.096 (1.514)	0.112 (2.260)	0.234 (3.847)	-0.008 (-0.205)	-0.028 (-0.525)
Lag positive	-0.014 (-0.188)	-0.032 (-0.429)	-0.041 (-0.686)	-0.020 (-0.346)	0.064 (0.827)	0.013 (0.228)	-0.111 (-1.692)	0.162 (2.622)	-0.158 (-2.525)
Lag collaboration	-0.088 (-2.125)	-0.077 (-1.739)	-0.082 (-2.117)	-0.121 (-3.159)	-0.067 (-1.517)	-0.036 (-1.049)	-0.018 (-0.364)	-0.127 (-3.429)	0.581 (10.136)
Output growth	0.063 (1.553)	0.045 (1.146)	0.059 (1.727)	-0.048 (-1.230)	-0.018 (-0.432)	-0.040 (-1.127)	-0.092 (-1.721)	0.051 (1.666)	0.034 (0.807)
Inflation	-0.060 (-1.119)	-0.019 (-0.366)	0.027 (0.564)	-0.016 (-0.287)	-0.029 (-0.428)	-0.019 (-0.338)	0.059 (0.867)	-0.036 (-0.998)	-0.013 (-0.245)
Unemployment	-0.084 (-0.420)	-0.286 (-1.634)	-0.149 (-0.906)	0.216 (1.151)	0.300 (1.278)	0.408 (2.035)	0.276 (1.058)	0.028 (0.199)	-0.026 (-0.120)
Housing starts	-1.067 (-1.680)	-0.275 (-0.460)	-0.012 (-0.023)	0.252 (0.424)	-0.882 (-1.435)	-0.116 (-0.228)	-0.606 (-0.996)	0.286 (0.595)	-0.646 (-0.987)
Lag output growth	-0.039 (-1.105)	-0.048 (-1.398)	-0.049 (-1.560)	0.054 (1.725)	0.009 (0.223)	0.053 (1.608)	0.097 (2.108)	-0.047 (-1.880)	-0.095 (-2.272)
Lag inflation	0.096 (1.719)	0.053 (0.989)	-0.014 (-0.287)	0.025 (0.436)	0.056 (0.770)	0.031 (0.515)	-0.048 (-0.670)	0.007 (0.183)	0.025 (0.450)
Lag unemployment	-0.024 (-0.117)	0.131 (0.732)	0.023 (0.136)	-0.188 (-1.035)	-0.369 (-1.634)	-0.499 (-2.513)	-0.317 (-1.212)	-0.054 (-0.383)	0.127 (0.578)
Lag Housing starts	0.784 (1.252)	0.471 (0.762)	0.302 (0.586)	-0.533 (-0.884)	0.757 (1.200)	0.276 (0.519)	0.284 (0.464)	0.101 (0.211)	0.716 (1.073)
R^2	0.243	0.407	0.533	0.128	0.244	0.351	0.308	0.198	0.467

Table A.4: Deliberation in Economic Round - Using Forecasts

Response Variable	Regressor				R^2
	Output Growth	Inflation	Unemployment	Housing Starts	
A. Tighter Stance					
Chair	-0.054 (-1.152)	0.027 (0.944)	0.018 (0.488)	-0.156 (-0.650)	0.019
Board	0.057 (1.118)	0.054 (1.817)	-0.268 (-6.479)	0.321 (1.469)	0.199
Bank	0.075 (1.596)	0.059 (1.941)	-0.314 (-8.687)	0.348 (1.558)	0.260
B. Looser Stance					
Chair	-0.046 (-1.148)	-0.029 (-1.188)	0.193 (4.256)	-0.381 (-1.631)	0.122
Board	0.032 (0.647)	-0.027 (-0.884)	0.177 (3.834)	-0.495 (-1.920)	0.117
Bank	0.064 (1.259)	-0.012 (-0.352)	0.136 (3.138)	-0.412 (-1.526)	0.069
C. Tone					
Negative	0.058 (0.974)	-0.024 (-0.634)	0.068 (1.451)	-0.229 (-0.854)	0.022
Positive	0.091 (2.670)	-0.010 (-0.418)	-0.121 (-2.860)	0.326 (1.658)	0.127
Collab.	-0.262 (-4.626)	0.016 (0.395)	0.168 (3.138)	-0.056 (-0.233)	0.166

Note: The figures in parenthesis are t statistics.

Table A.5: Deliberation in Policy Round - Using Forecasts

Regressor	Response Variable								
	Tight	Tight	Tight	Loose	Loose	Loose	Tone		
	Chair	Board	Bank	Chair	Board	Bank	Negative	Positive	Collab.
Intercept	1.112 (2.705)	1.757 (4.205)	2.209 (5.216)	-0.016 (-0.043)	0.695 (1.562)	0.839 (2.409)	0.219 (0.563)	-0.292 (-0.859)	-1.623 (-3.327)
Lag tight chair	-0.093 (-1.680)	0.054 (0.886)	0.015 (0.221)	-0.096 (-1.407)	-0.037 (-0.725)	0.036 (0.744)	0.075 (1.317)	-0.021 (-0.421)	0.115 (1.642)
Lag tight board	0.336 (4.215)	0.251 (2.955)	0.145 (1.821)	-0.159 (-2.252)	-0.103 (-1.447)	-0.073 (-1.234)	-0.098 (-1.150)	0.057 (0.790)	-0.202 (-1.992)
Lag tight bank	-0.118 (-1.331)	-0.087 (-0.856)	0.026 (0.259)	0.062 (0.775)	-0.134 (-1.764)	-0.129 (-1.880)	-0.042 (-0.432)	-0.151 (-1.670)	0.320 (3.052)
Lag loose chair	-0.064 (-1.037)	-0.067 (-1.232)	0.030 (0.502)	-0.019 (-0.414)	0.123 (2.055)	0.115 (2.115)	0.061 (1.364)	-0.067 (-1.407)	-0.075 (-1.510)
Lag loose board	0.049 (0.761)	-0.052 (-0.822)	-0.085 (-1.272)	-0.036 (-0.576)	-0.011 (-0.169)	0.087 (1.084)	-0.049 (-0.716)	0.009 (0.182)	-0.134 (-1.657)
Lag loose bank	-0.165 (-2.459)	-0.125 (-1.803)	-0.217 (-3.105)	0.125 (1.730)	0.110 (1.873)	0.183 (3.474)	0.084 (1.417)	0.028 (0.689)	0.211 (3.060)
Lag negative	-0.060 (-1.069)	-0.069 (-1.103)	-0.050 (-0.879)	0.045 (0.635)	0.080 (1.345)	0.018 (0.332)	0.008 (0.145)	-0.075 (-1.507)	-0.020 (-0.310)
Lag positive	-0.088 (-1.138)	-0.025 (-0.382)	0.066 (0.943)	0.043 (0.599)	-0.029 (-0.510)	0.026 (0.400)	-0.042 (-0.661)	0.194 (1.730)	-0.074 (-0.837)
Lag collaboration	0.052 (0.992)	0.081 (1.432)	0.019 (0.357)	-0.045 (-1.030)	0.016 (0.351)	0.037 (0.884)	-0.025 (-0.474)	-0.042 (-1.133)	0.313 (4.766)
Output growth	0.101 (2.157)	-0.033 (-0.673)	-0.069 (-1.444)	-0.015 (-0.362)	0.030 (0.663)	0.012 (0.302)	-0.004 (-0.084)	0.088 (2.141)	-0.149 (-2.663)
Inflation	0.079 (2.776)	0.079 (2.513)	0.089 (2.920)	0.015 (0.552)	0.035 (1.358)	0.035 (1.469)	-0.001 (-0.042)	0.008 (0.298)	-0.089 (-2.612)
Unemployment	-0.186 (-3.895)	-0.207 (-4.359)	-0.179 (-3.912)	-0.016 (-0.413)	-0.065 (-1.380)	-0.056 (-1.400)	0.014 (0.295)	-0.059 (-1.602)	0.241 (4.408)
Housing starts	-0.329 (-1.484)	-0.421 (-1.880)	-0.811 (-3.617)	0.079 (0.361)	-0.336 (-1.433)	-0.439 (-2.181)	-0.204 (-0.989)	0.277 (1.212)	0.554 (2.261)
R^2	0.198	0.220	0.263	0.053	0.149	0.236	0.061	0.134	0.213

Note: The figures in parenthesis are t statistics.

Table A.6: Deliberation Across Meetings - Using Forecasts

Regressor	Response Variable								
	Tight	Tight	Tight	Loose	Loose	Loose	Tone		
	Chair	Board	Bank	Chair	Board	Bank	Negative	Positive	Collab.
Intercept	1.052 (2.539)	0.841 (2.101)	0.954 (2.640)	-0.301 (-0.773)	0.412 (0.993)	0.238 (0.623)	0.896 (2.328)	0.130 (0.347)	-0.853 (-2.209)
Lag tight chair	0.117 (1.742)	-0.010 (-0.161)	-0.030 (-0.627)	0.067 (0.913)	-0.024 (-0.451)	-0.018 (-0.334)	-0.004 (-0.060)	-0.046 (-0.931)	-0.042 (-1.006)
Lag tight board	0.085 (1.049)	0.308 (4.127)	0.082 (1.252)	-0.095 (-1.184)	-0.076 (-1.179)	-0.008 (-0.124)	0.088 (1.102)	0.020 (0.300)	0.101 (1.311)
Lag tight bank	0.011 (0.128)	0.027 (0.332)	0.182 (2.507)	0.017 (0.204)	-0.205 (-2.674)	-0.191 (-2.826)	-0.201 (-2.342)	-0.002 (-0.041)	0.099 (1.341)
Lag loose chair	0.070 (1.078)	0.085 (1.551)	0.039 (0.895)	0.137 (2.075)	0.011 (0.208)	0.024 (0.514)	0.016 (0.307)	0.010 (0.221)	-0.046 (-0.910)
Lag loose board	-0.190 (-2.773)	-0.026 (-0.377)	-0.066 (-1.233)	0.095 (1.539)	0.059 (0.813)	0.116 (1.466)	0.050 (0.557)	-0.026 (-0.486)	0.087 (1.522)
Lag loose bank	-0.070 (-0.931)	-0.173 (-2.701)	-0.190 (-3.788)	-0.050 (-0.666)	0.149 (1.918)	0.313 (5.297)	0.123 (1.464)	0.049 (0.938)	0.028 (0.472)
Lag negative	0.043 (0.653)	-0.016 (-0.276)	-0.118 (-2.372)	0.136 (1.969)	0.086 (1.656)	0.181 (3.562)	0.214 (3.456)	0.002 (0.042)	-0.005 (-0.095)
Lag positive	0.057 (0.793)	0.010 (0.129)	0.011 (0.176)	0.041 (0.643)	0.110 (1.310)	-0.023 (-0.386)	-0.128 (-1.960)	0.132 (2.056)	-0.152 (-2.428)
Lag collaboration	-0.053 (-1.071)	-0.041 (-0.862)	-0.076 (-1.812)	-0.086 (-2.038)	-0.091 (-2.169)	-0.000 (-0.013)	-0.019 (-0.370)	-0.089 (-2.298)	0.577 (10.415)
Output growth	0.136 (1.814)	-0.007 (-0.091)	-0.004 (-0.067)	-0.035 (-0.422)	-0.115 (-1.424)	0.076 (1.044)	-0.098 (-1.173)	0.100 (1.754)	-0.063 (-0.722)
Inflation	-0.022 (-0.172)	0.136 (1.160)	0.193 (1.613)	0.069 (0.512)	-0.062 (-0.512)	0.067 (0.520)	-0.150 (-0.968)	0.002 (0.017)	-0.031 (-0.211)
Unemployment	-0.217 (-1.045)	-0.530 (-3.087)	-0.561 (-3.349)	0.463 (2.487)	1.081 (5.417)	0.710 (3.863)	0.863 (3.720)	-0.286 (-2.004)	-0.315 (-1.697)
Housing starts	-1.786 (-2.547)	-0.073 (-0.104)	-0.586 (-0.895)	0.440 (0.557)	0.164 (0.218)	-0.979 (-1.274)	-0.598 (-0.770)	0.451 (0.786)	-0.608 (-0.972)
Lag output growth	-0.100 (-1.372)	0.001 (0.017)	-0.010 (-0.173)	0.011 (0.139)	0.137 (1.792)	-0.022 (-0.324)	0.133 (1.719)	-0.005 (-0.084)	-0.058 (-0.674)
Lag inflation	0.099 (0.768)	-0.075 (-0.650)	-0.135 (-1.152)	-0.075 (-0.538)	0.091 (0.812)	-0.036 (-0.281)	0.176 (1.160)	0.009 (0.094)	-0.003 (-0.019)
Lag unemployment	0.087 (0.423)	0.353 (2.102)	0.372 (2.190)	-0.394 (-2.180)	-1.075 (-5.447)	-0.770 (-4.316)	-0.918 (-3.934)	0.177 (1.255)	0.477 (2.470)
Lag housing starts	1.386 (1.972)	0.148 (0.210)	0.658 (0.977)	-0.473 (-0.606)	-0.580 (-0.790)	0.913 (1.105)	0.092 (0.123)	-0.269 (-0.472)	0.794 (1.246)
R^2	0.200	0.408	0.518	0.143	0.373	0.487	0.328	0.222	0.483

Table A.7: Deliberation in Economic Round - Break in Oct. 1993

Response Variable	Regressor				R^2
	Output Growth	Inflation	Unemployment	Housing Starts	
A. Tighter Stance					
Chair	0.014 (0.509)	0.032 (1.443)	-0.047 (-1.378)	-0.288 (-1.504)	0.010
Board	0.089 (3.105)	0.025 (0.964)	-0.165 (-4.417)	0.143 (0.769)	0.142
Bank	0.140 (4.536)	0.023 (0.850)	-0.178 (-5.461)	0.166 (0.833)	0.229
B. Looser Stance					
Chair	-0.041 (-1.580)	-0.028 (-1.547)	0.147 (3.497)	-0.253 (-1.296)	0.095
Board	-0.132 (-4.238)	-0.053 (-2.493)	0.162 (3.827)	0.076 (0.401)	0.131
Bank	-0.184 (-4.958)	-0.062 (-1.997)	0.150 (3.621)	0.441 (1.817)	0.129
C. Tone					
Negative	-0.152 (-4.542)	-0.079 (-2.841)	0.108 (2.436)	0.547 (2.551)	0.073
Positive	0.017 (0.785)	-0.040 (-2.157)	-0.033 (-0.858)	0.448 (2.658)	0.080
Collab.	-0.084 (-2.389)	0.073 (1.917)	0.033 (0.680)	-0.167 (-0.747)	0.085

Note: The figures in parenthesis are t statistics.

Table A.8: Deliberation in Policy Round - Break in Oct. 1993

Regressor	Response Variable								
	Tight	Tight	Tight	Loose	Loose	Loose	Tone		
	Chair	Board	Bank	Chair	Board	Bank	Negative	Positive	Collab.
Intercept	0.519 (1.398)	1.778 (4.534)	2.187 (5.250)	-0.139 (-0.401)	0.710 (1.742)	0.955 (3.009)	-0.023 (-0.063)	-0.355 (-1.184)	-1.883 (-4.277)
Lag tight chair	-0.086 (-1.533)	0.070 (1.121)	0.059 (0.843)	-0.079 (-1.118)	-0.035 (-0.670)	0.061 (1.236)	0.085 (1.468)	-0.028 (-0.548)	0.049 (0.697)
Lag tight board	0.285 (3.417)	0.239 (2.797)	0.128 (1.618)	-0.194 (-2.934)	-0.108 (-1.517)	-0.096 (-1.548)	-0.094 (-1.071)	0.030 (0.441)	-0.138 (-1.446)
Lag tight bank	-0.063 (-0.705)	-0.084 (-0.802)	0.010 (0.104)	0.090 (1.203)	-0.136 (-1.864)	-0.148 (-2.262)	-0.057 (-0.618)	-0.138 (-1.659)	0.383 (3.945)
Lag loose chair	-0.058 (-0.959)	-0.070 (-1.299)	0.042 (0.716)	-0.008 (-0.166)	0.123 (2.021)	0.122 (2.178)	0.076 (1.685)	-0.087 (-1.867)	-0.054 (-1.163)
Lag loose board	0.017 (0.247)	-0.075 (-1.176)	-0.099 (-1.517)	-0.053 (-0.876)	-0.016 (-0.244)	0.075 (0.973)	-0.037 (-0.544)	-0.015 (-0.319)	-0.081 (-1.123)
Lag loose bank	-0.124 (-1.809)	-0.140 (-1.980)	-0.234 (-3.340)	0.135 (1.917)	0.118 (2.054)	0.183 (3.591)	0.078 (1.270)	0.056 (1.357)	0.171 (2.636)
Lag negative	-0.009 (-0.144)	-0.074 (-1.161)	-0.042 (-0.737)	0.075 (1.076)	0.086 (1.389)	0.033 (0.596)	0.013 (0.223)	-0.061 (-1.230)	-0.078 (-1.313)
Lag positive	-0.073 (-0.940)	0.015 (0.233)	0.077 (1.116)	0.030 (0.431)	-0.012 (-0.221)	0.049 (0.797)	-0.062 (-1.022)	0.239 (2.243)	-0.185 (-2.237)
Lag collaboration	0.059 (1.143)	0.061 (1.126)	0.040 (0.792)	-0.026 (-0.655)	0.012 (0.258)	0.035 (0.856)	-0.012 (-0.233)	-0.073 (-1.923)	0.339 (5.664)
Output growth	-0.018 (-0.701)	0.032 (1.077)	0.045 (1.537)	-0.046 (-1.779)	0.032 (1.348)	0.030 (1.333)	0.014 (0.482)	-0.016 (-0.658)	-0.107 (-3.705)
Inflation	0.036 (1.644)	0.088 (3.396)	0.097 (3.569)	0.015 (0.698)	0.032 (1.548)	0.052 (2.957)	0.008 (0.310)	-0.010 (-0.508)	-0.077 (-2.969)
Unemployment	-0.089 (-2.328)	-0.217 (-5.239)	-0.215 (-5.027)	0.005 (0.147)	-0.066 (-1.627)	-0.078 (-2.316)	0.013 (0.309)	-0.006 (-0.217)	0.208 (4.388)
Housingstarts	-0.023 (-0.122)	-0.554 (-2.827)	-0.901 (-4.236)	0.117 (0.617)	-0.338 (-1.730)	-0.506 (-3.163)	-0.082 (-0.437)	0.335 (1.826)	0.770 (3.903)
R^2	0.119	0.227	0.253	0.083	0.138	0.242	0.055	0.115	0.206

Note: The figures in parenthesis are t statistics.

Table A.9: Deliberation Across Meetings - Break in Oct. 1993

Regressor	Response Variable								
	Tight	Tight	Tight	Loose	Loose	Loose	Tone		
	Chair	Board	Bank	Chair	Board	Bank	Negative	Positive	Collab.
Intercept	0.971 (2.435)	0.786 (2.138)	0.607 (1.807)	-0.347 (-0.870)	0.193 (0.535)	0.221 (0.672)	0.373 (0.950)	-0.248 (-0.714)	-0.885 (-2.455)
Lag tight chair	0.101 (1.477)	-0.037 (-0.600)	-0.059 (-1.190)	0.059 (0.802)	0.001 (0.020)	0.010 (0.177)	0.039 (0.585)	-0.050 (-0.977)	-0.029 (-0.638)
Lag tight board	0.077 (0.949)	0.311 (4.102)	0.093 (1.426)	-0.092 (-1.138)	-0.093 (-1.377)	-0.038 (-0.582)	0.068 (0.824)	0.022 (0.312)	0.069 (0.883)
Lag tight bank	-0.012 (-0.137)	0.005 (0.067)	0.148 (2.122)	-0.004 (-0.045)	-0.164 (-2.368)	-0.175 (-2.678)	-0.166 (-1.910)	-0.034 (-0.591)	0.170 (2.198)
Lag loose chair	0.042 (0.639)	0.055 (0.984)	0.006 (0.131)	0.133 (2.005)	0.017 (0.300)	0.028 (0.595)	0.039 (0.709)	-0.013 (-0.266)	-0.019 (-0.385)
Lag loose board	-0.186 (-2.601)	-0.026 (-0.353)	-0.053 (-0.906)	0.102 (1.620)	0.064 (0.835)	0.098 (1.227)	0.046 (0.493)	-0.036 (-0.676)	0.083 (1.413)
Lag loose bank	-0.063 (-0.804)	-0.174 (-2.629)	-0.186 (-3.799)	-0.050 (-0.647)	0.148 (1.852)	0.317 (5.306)	0.126 (1.505)	0.052 (0.994)	-0.012 (-0.213)
Lag negative	0.044 (0.703)	-0.043 (-0.757)	-0.140 (-2.871)	0.149 (2.169)	0.141 (2.655)	0.204 (4.157)	0.259 (4.204)	-0.028 (-0.693)	-0.006 (-0.108)
Lag positive	0.042 (0.570)	0.019 (0.245)	-0.010 (-0.158)	0.028 (0.442)	0.106 (1.185)	-0.003 (-0.050)	-0.132 (-1.916)	0.160 (2.509)	-0.189 (-3.127)
Lag collaboration	-0.012 (-0.235)	-0.034 (-0.780)	-0.032 (-0.793)	-0.038 (-0.827)	-0.079 (-1.712)	-0.011 (-0.282)	-0.010 (-0.206)	-0.127 (-3.429)	0.566 (10.365)
Output growth	0.051 (1.106)	0.054 (1.526)	0.068 (1.785)	-0.117 (-2.742)	-0.023 (-0.547)	-0.073 (-2.106)	-0.092 (-1.778)	0.045 (1.516)	0.040 (0.915)
Inflation	-0.044 (-0.716)	-0.031 (-0.558)	0.032 (0.633)	0.031 (0.588)	-0.008 (-0.115)	-0.004 (-0.066)	0.059 (0.851)	-0.025 (-0.682)	-0.028 (-0.522)
Unemployment	-0.048 (-0.227)	-0.406 (-2.459)	-0.284 (-1.666)	0.251 (1.420)	0.402 (1.757)	0.322 (1.767)	0.336 (1.287)	-0.004 (-0.030)	0.049 (0.223)
Housingstarts	-1.179 (-1.798)	-0.191 (-0.339)	-0.607 (-1.195)	0.211 (0.344)	-0.821 (-1.381)	0.247 (0.500)	-0.518 (-0.866)	0.221 (0.472)	-0.496 (-0.824)
Lag output growth	-0.038 (-0.955)	-0.044 (-1.262)	-0.048 (-1.347)	0.089 (2.564)	-0.003 (-0.074)	0.064 (2.080)	0.089 (1.986)	-0.043 (-1.708)	-0.105 (-2.486)
Lag inflation	0.100 (1.619)	0.087 (1.528)	0.005 (0.091)	-0.039 (-0.682)	0.026 (0.355)	0.017 (0.291)	-0.060 (-0.817)	0.003 (0.074)	0.021 (0.377)
Lag unemployment	-0.070 (-0.338)	0.253 (1.523)	0.139 (0.813)	-0.189 (-1.122)	-0.394 (-1.776)	-0.368 (-2.034)	-0.356 (-1.371)	-0.022 (-0.151)	0.067 (0.307)
Lag housingstarts	0.864 (1.281)	0.177 (0.301)	0.714 (1.411)	-0.177 (-0.273)	0.654 (1.116)	-0.207 (-0.402)	0.346 (0.577)	0.127 (0.270)	0.728 (1.195)
R^2	0.143	0.353	0.436	0.170	0.317	0.472	0.292	0.168	0.429

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